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(54) **METHOD FOR MAKING FIBRES**

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(57)

ABSTRACT

This invention relates to a method for making fibres, the method comprising: providing a spinning dope comprising lignin and an additive polymer dissolved in a dope solvent, wherein the dope solvent comprises an ionic liquid and water; and extruding the spinning dope into a coagulation bath to obtain one or more fibres.

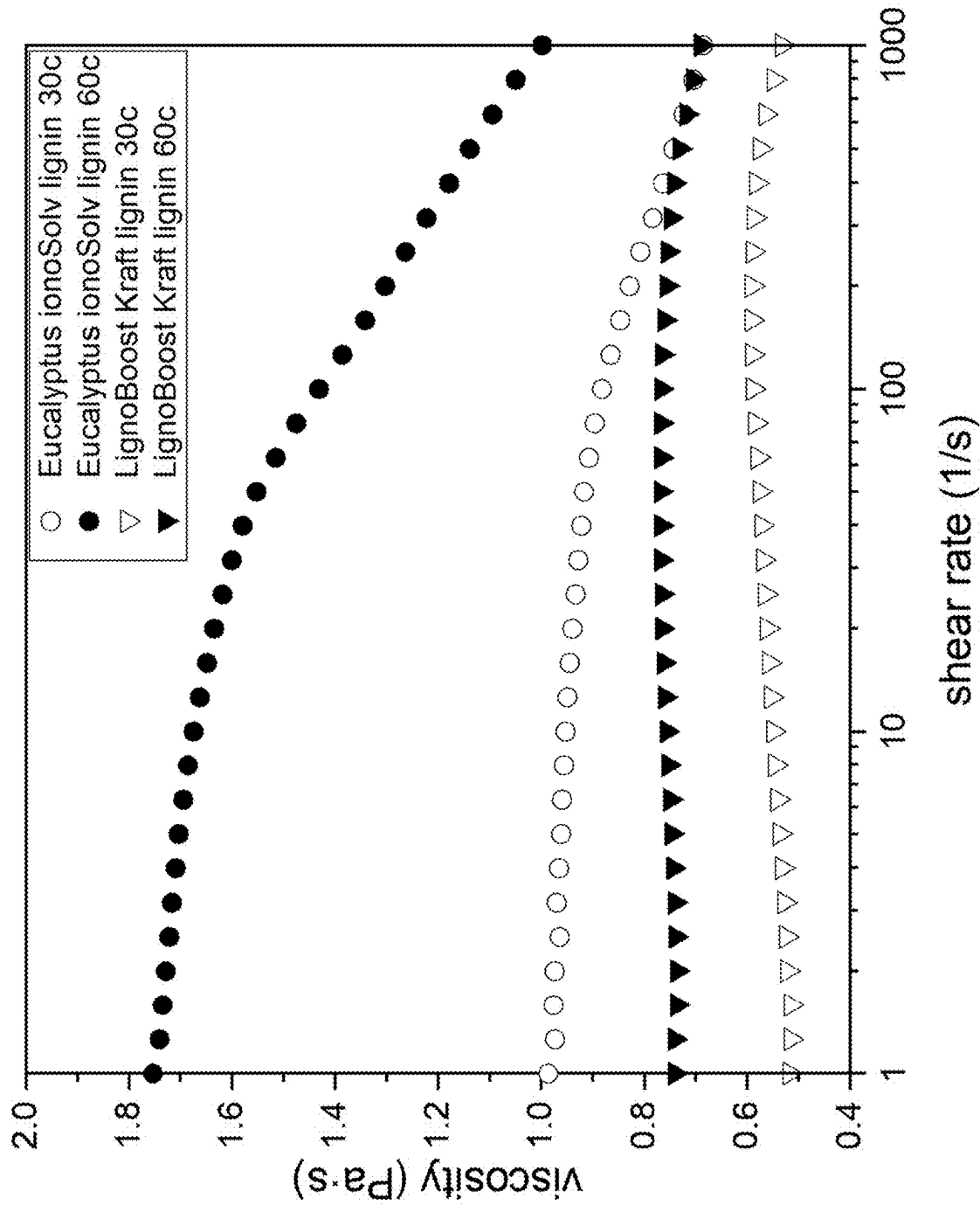


FIG. 1

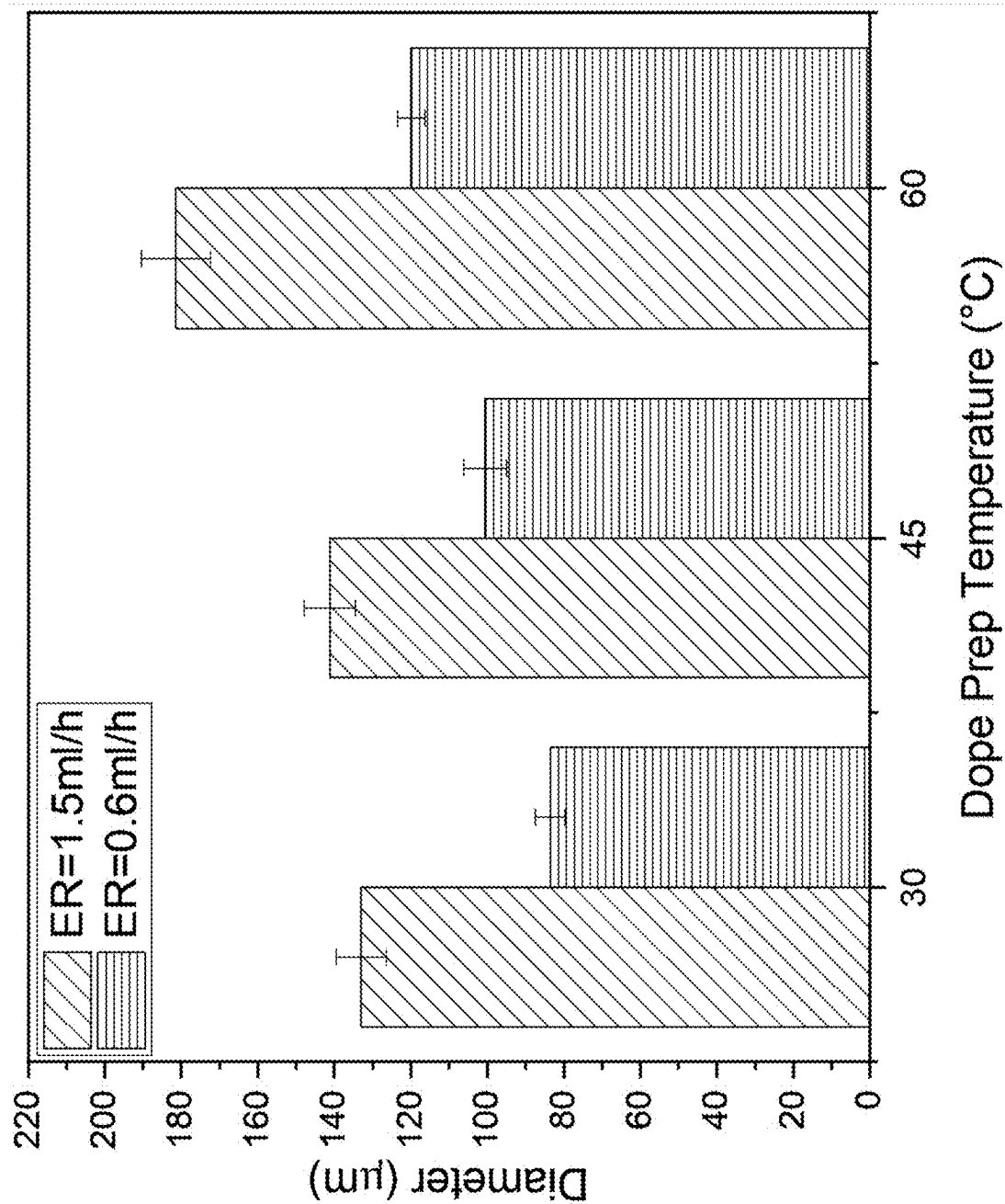


FIG. 2

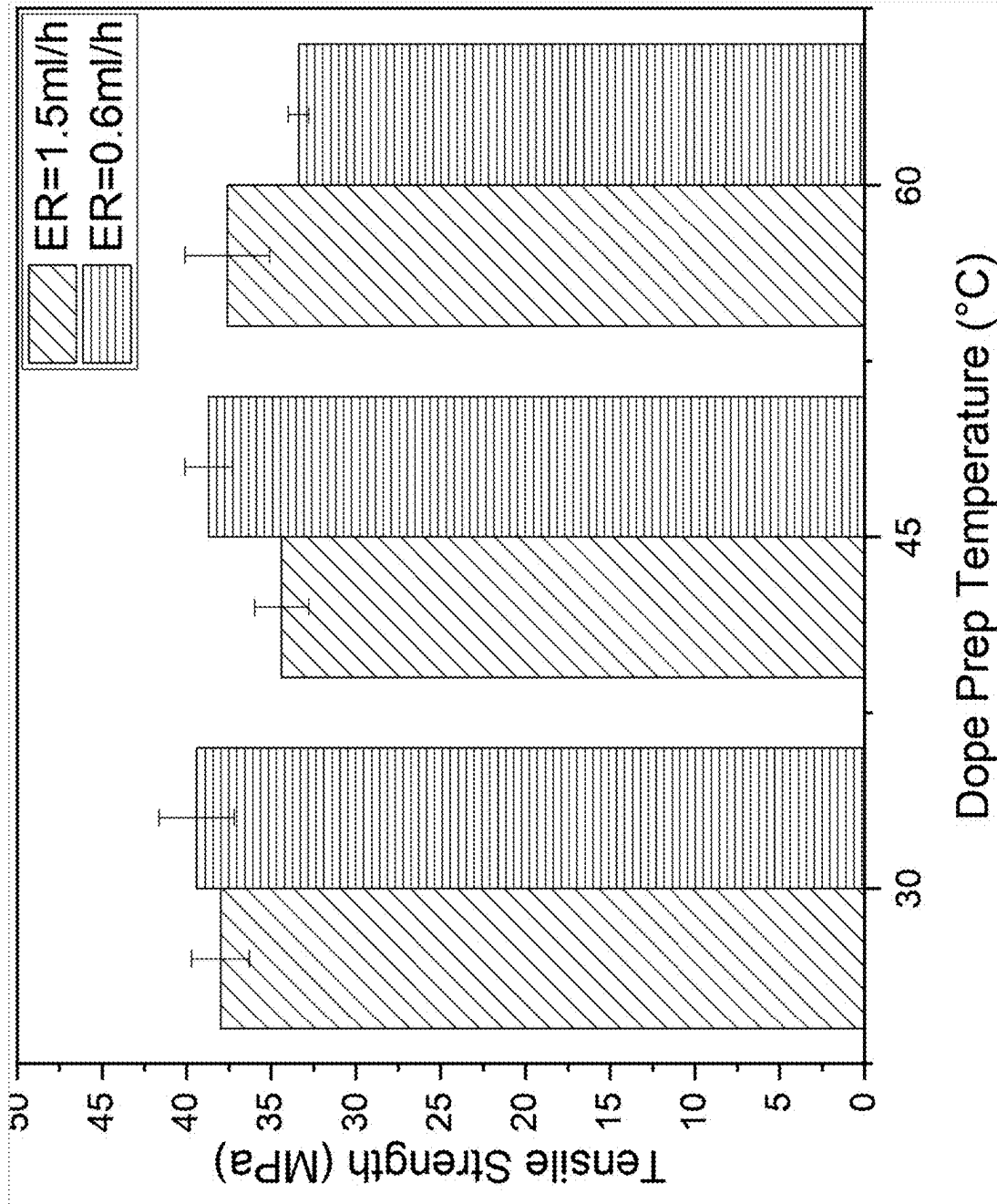


FIG. 3a

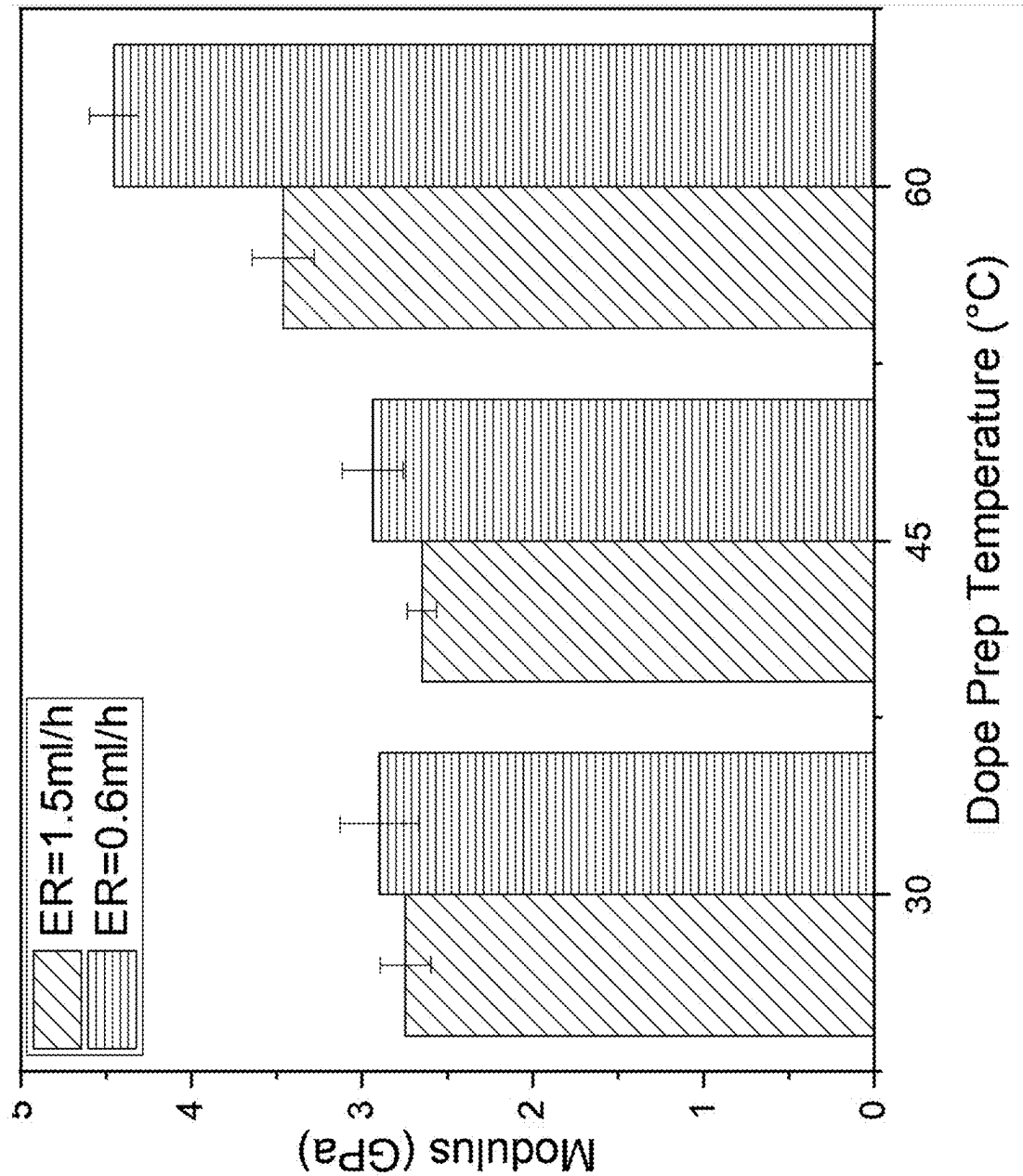


FIG. 3b

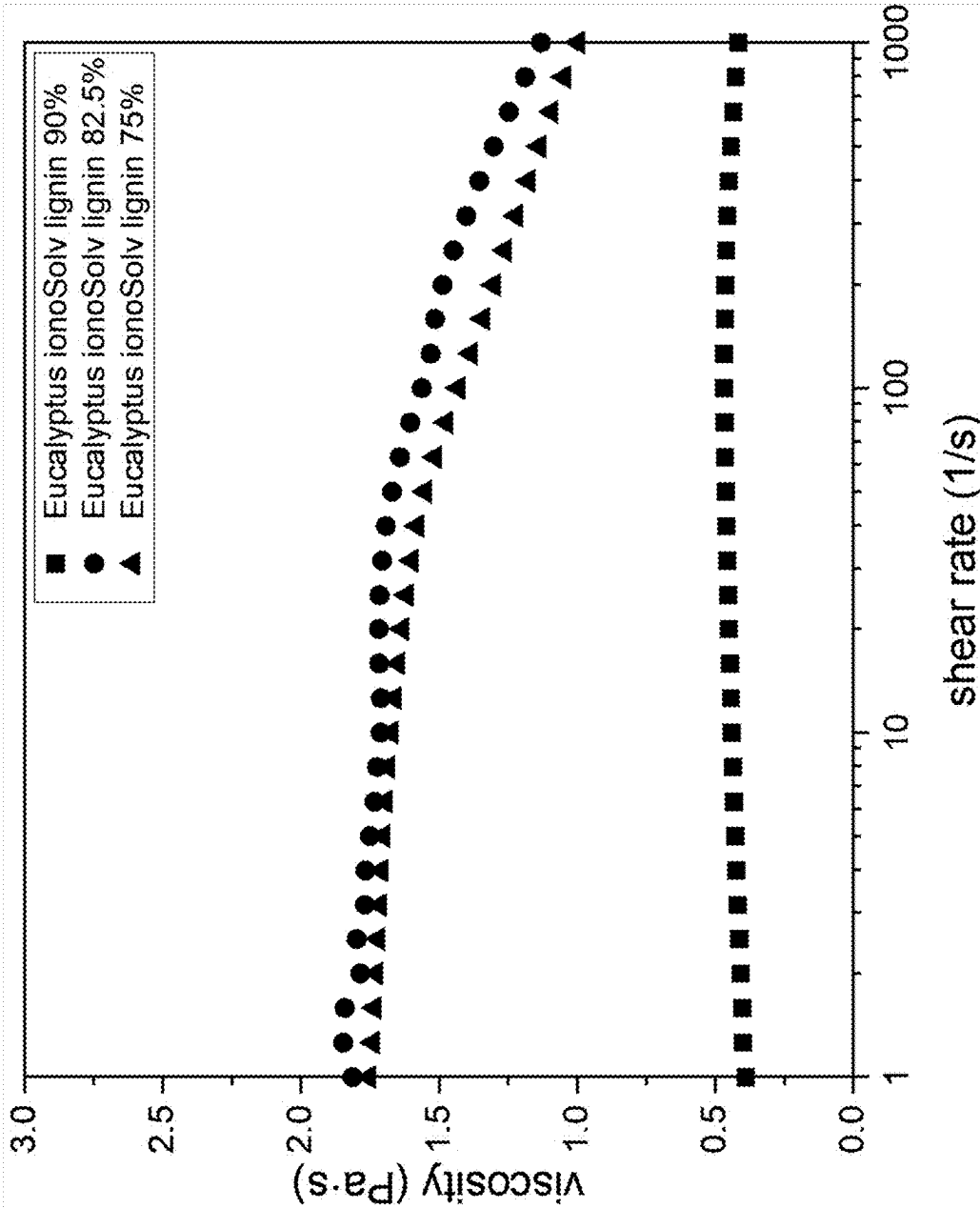


FIG. 4a

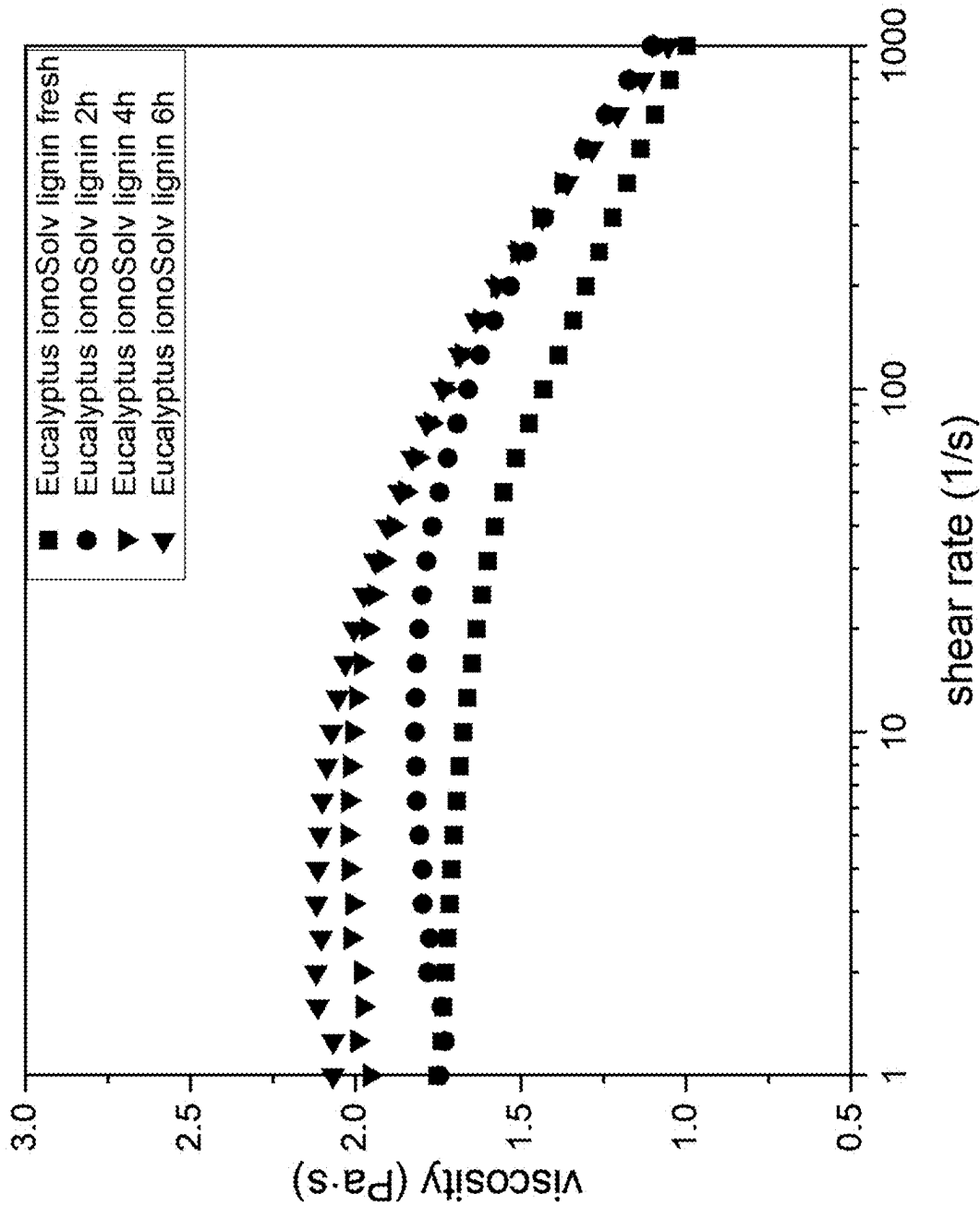


FIG. 4b

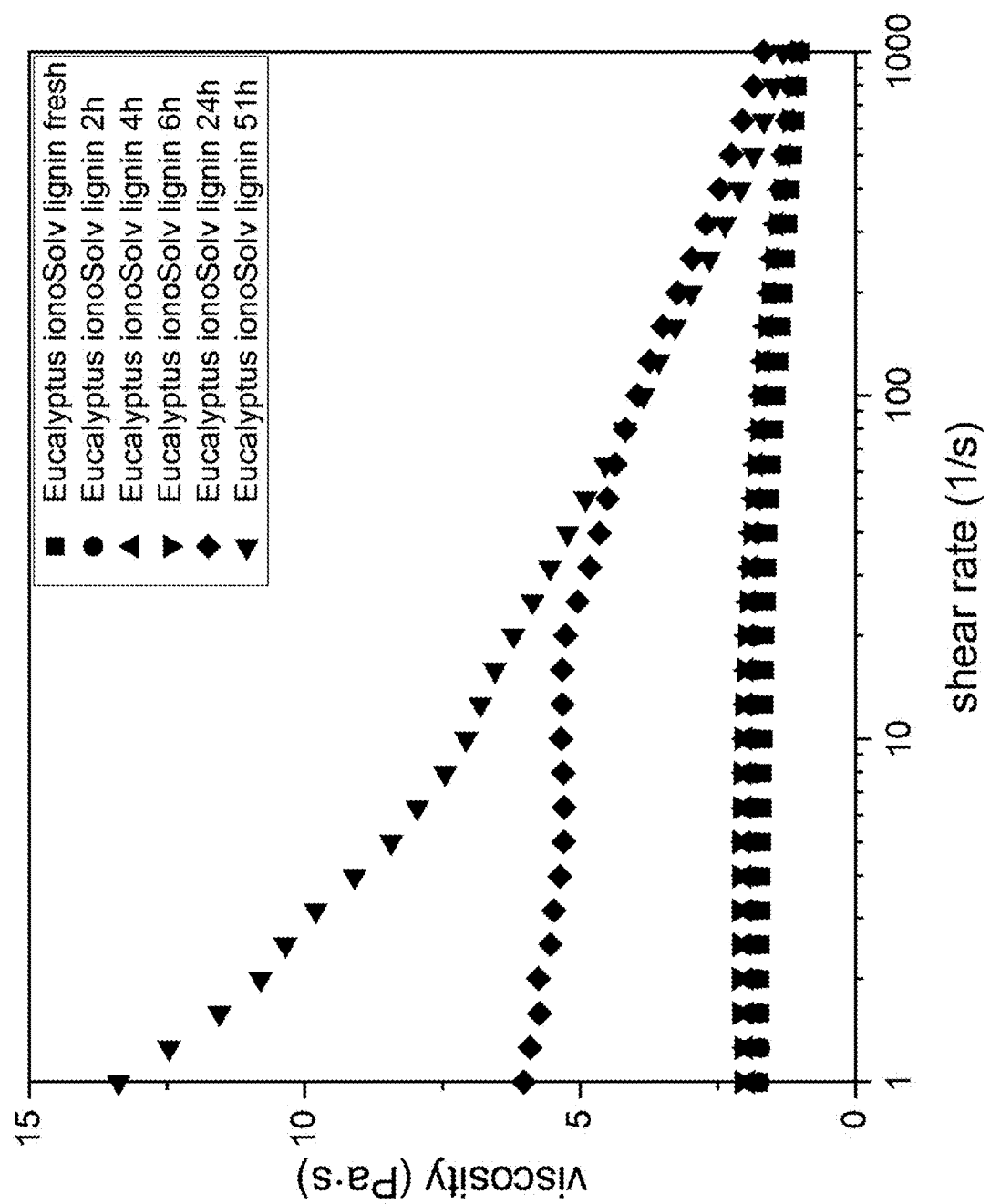


FIG. 5

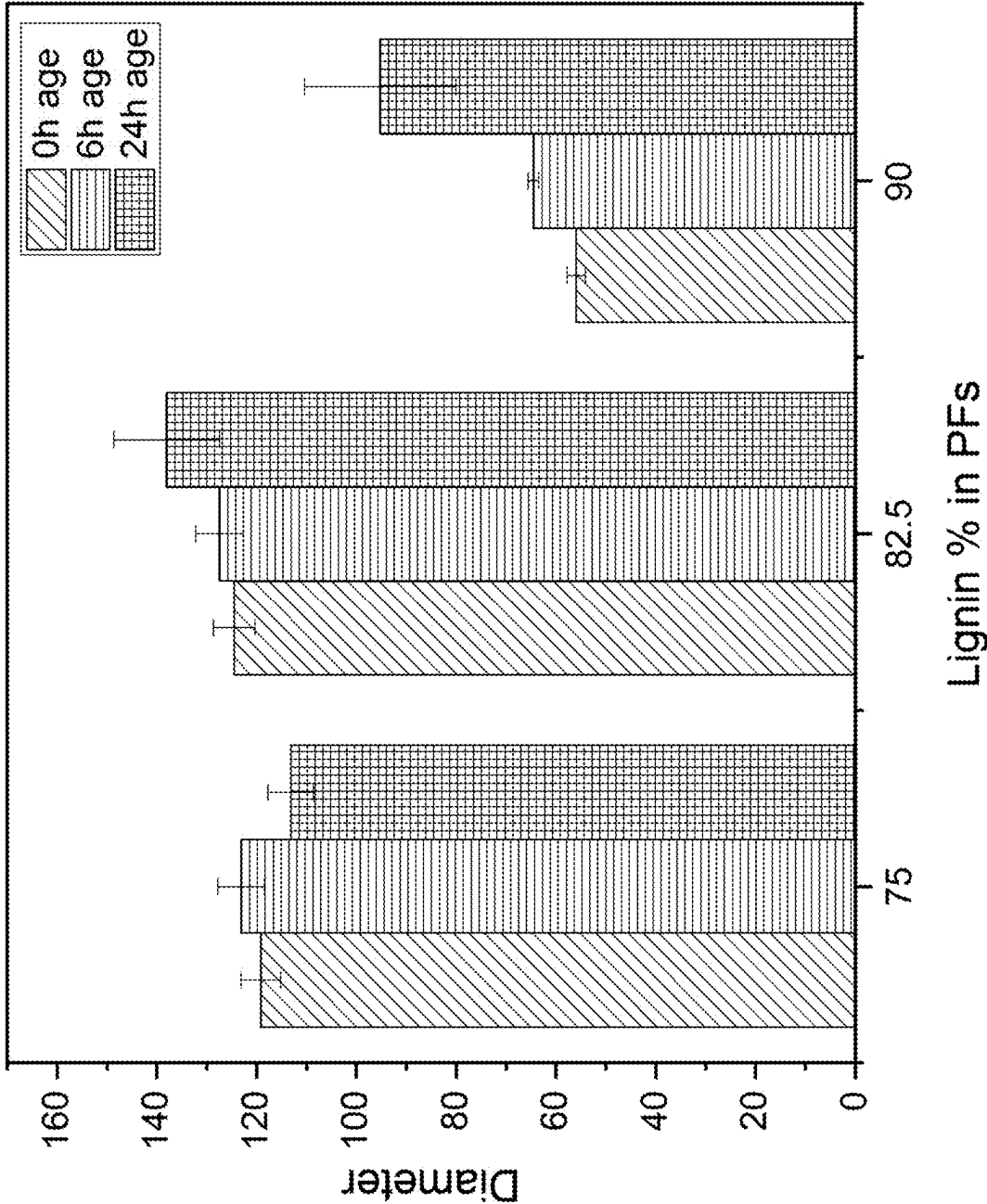


FIG. 6a

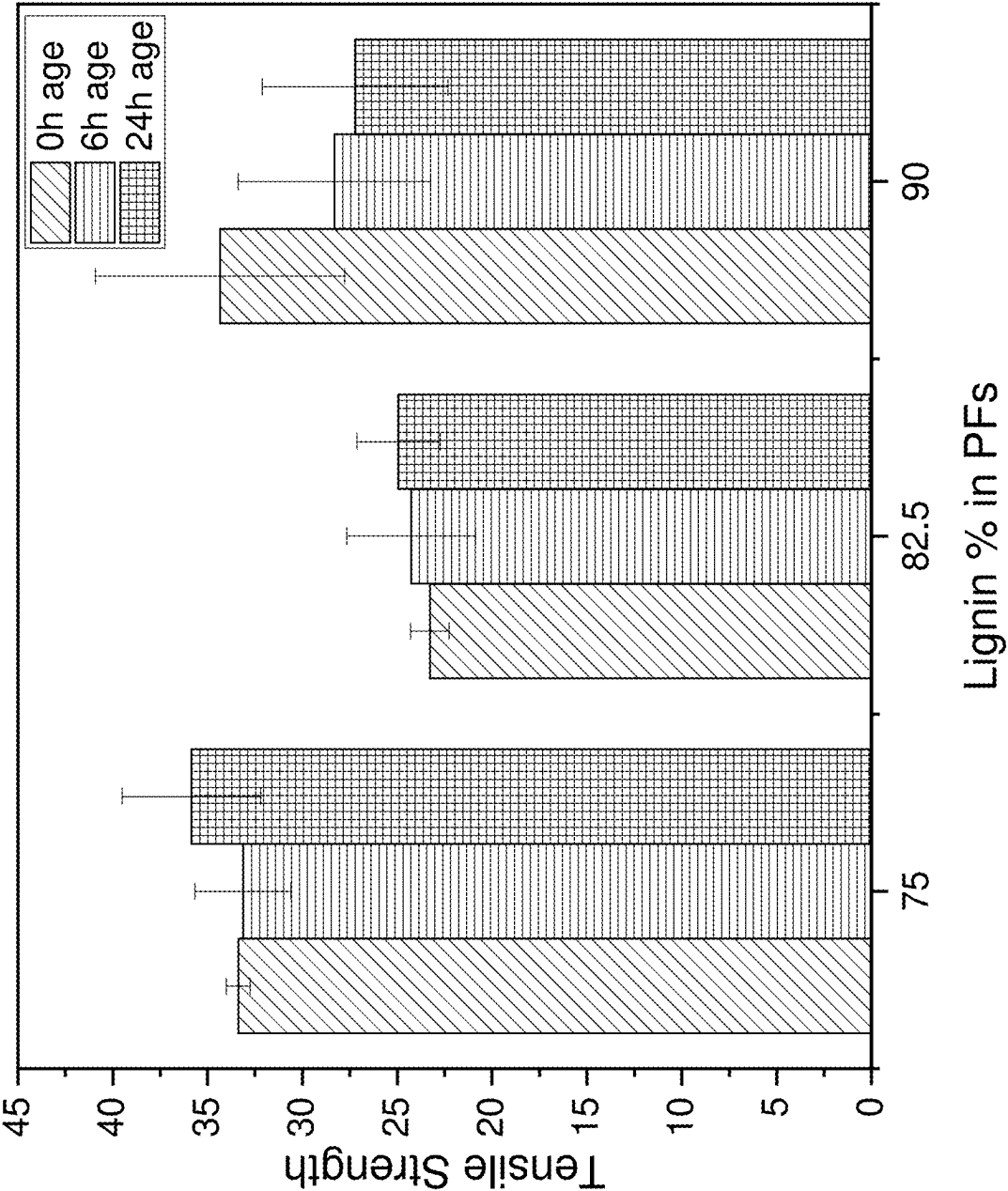


FIG. 6b

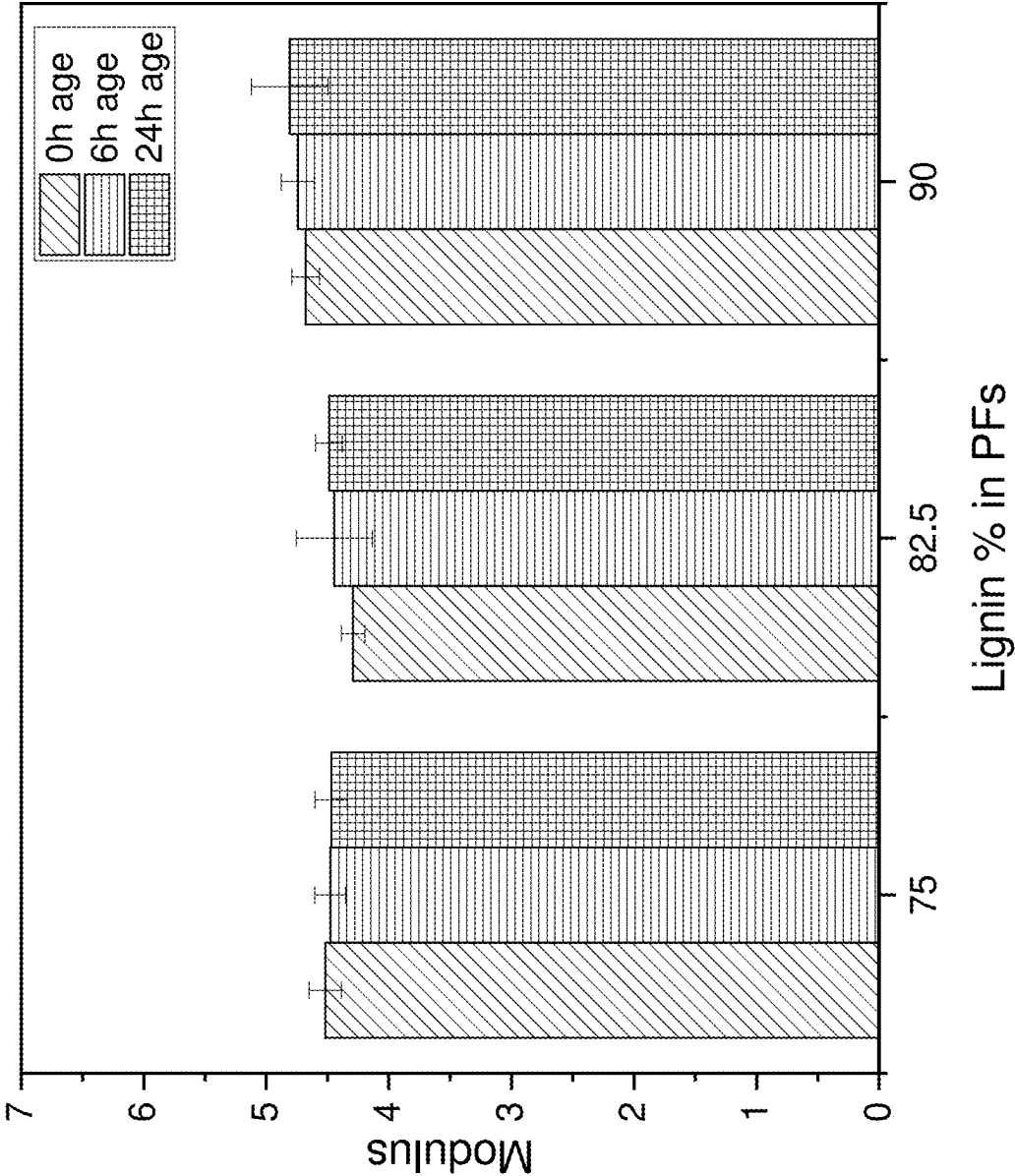


FIG. 6c

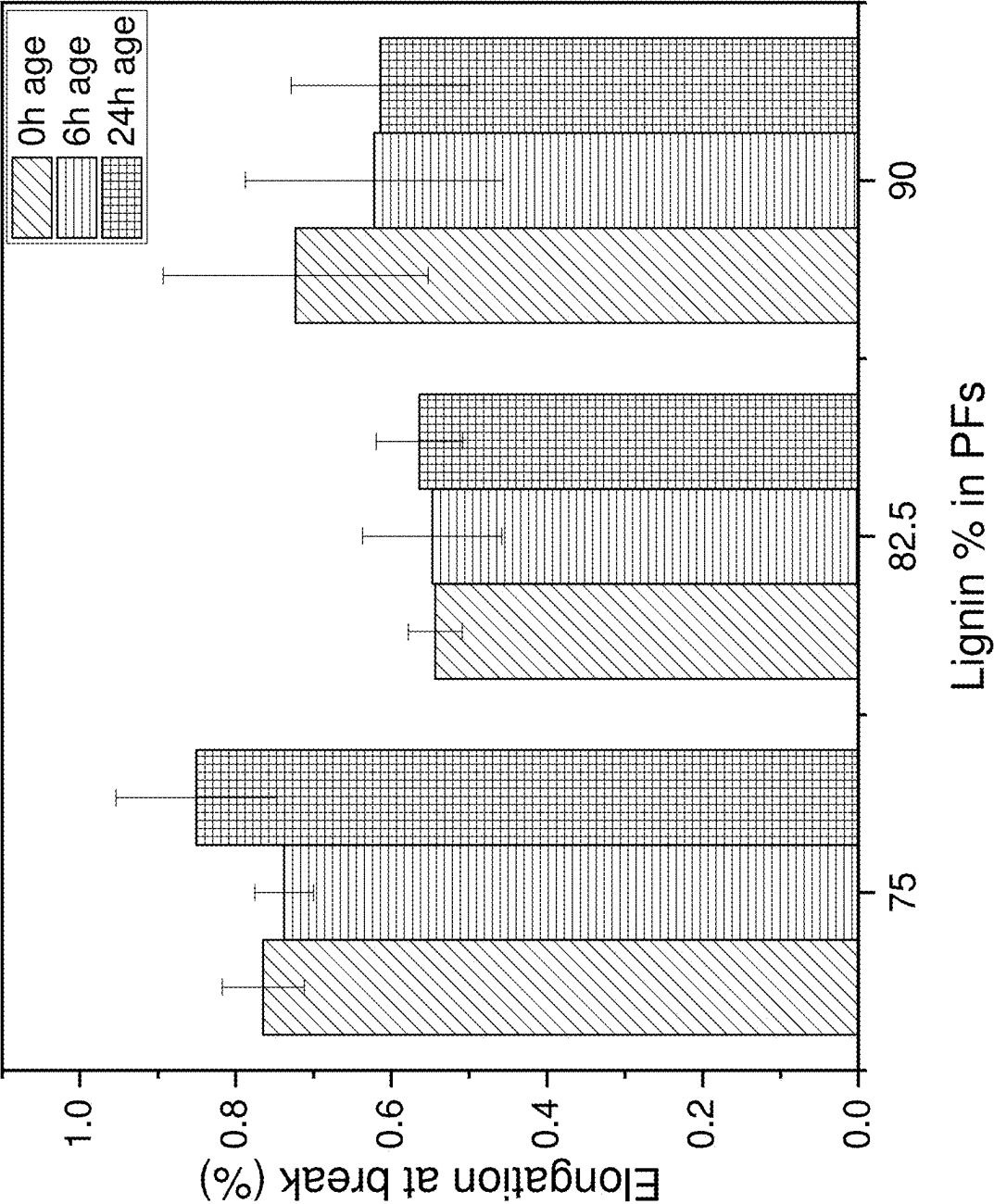


FIG. 6d

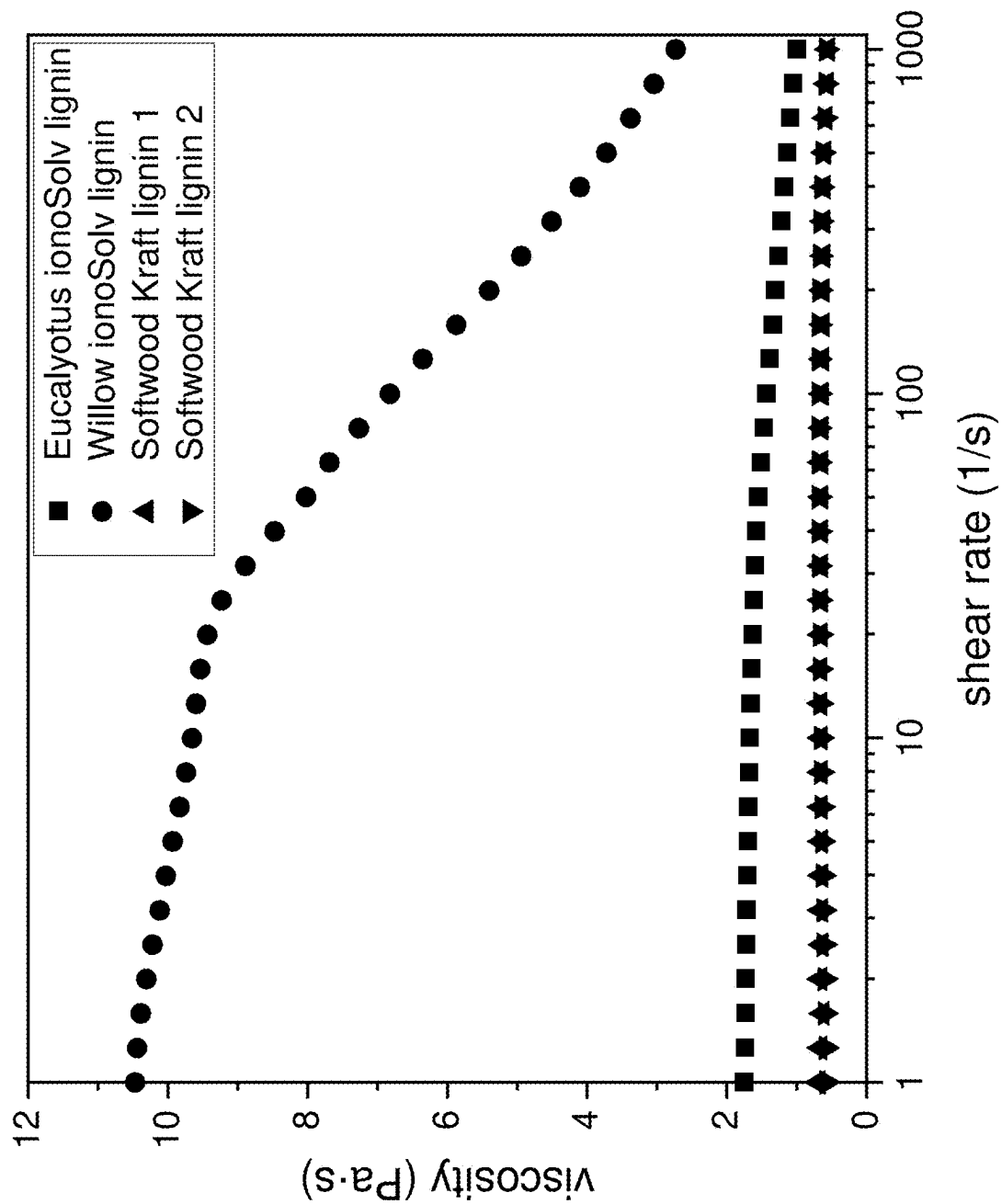


FIG. 7

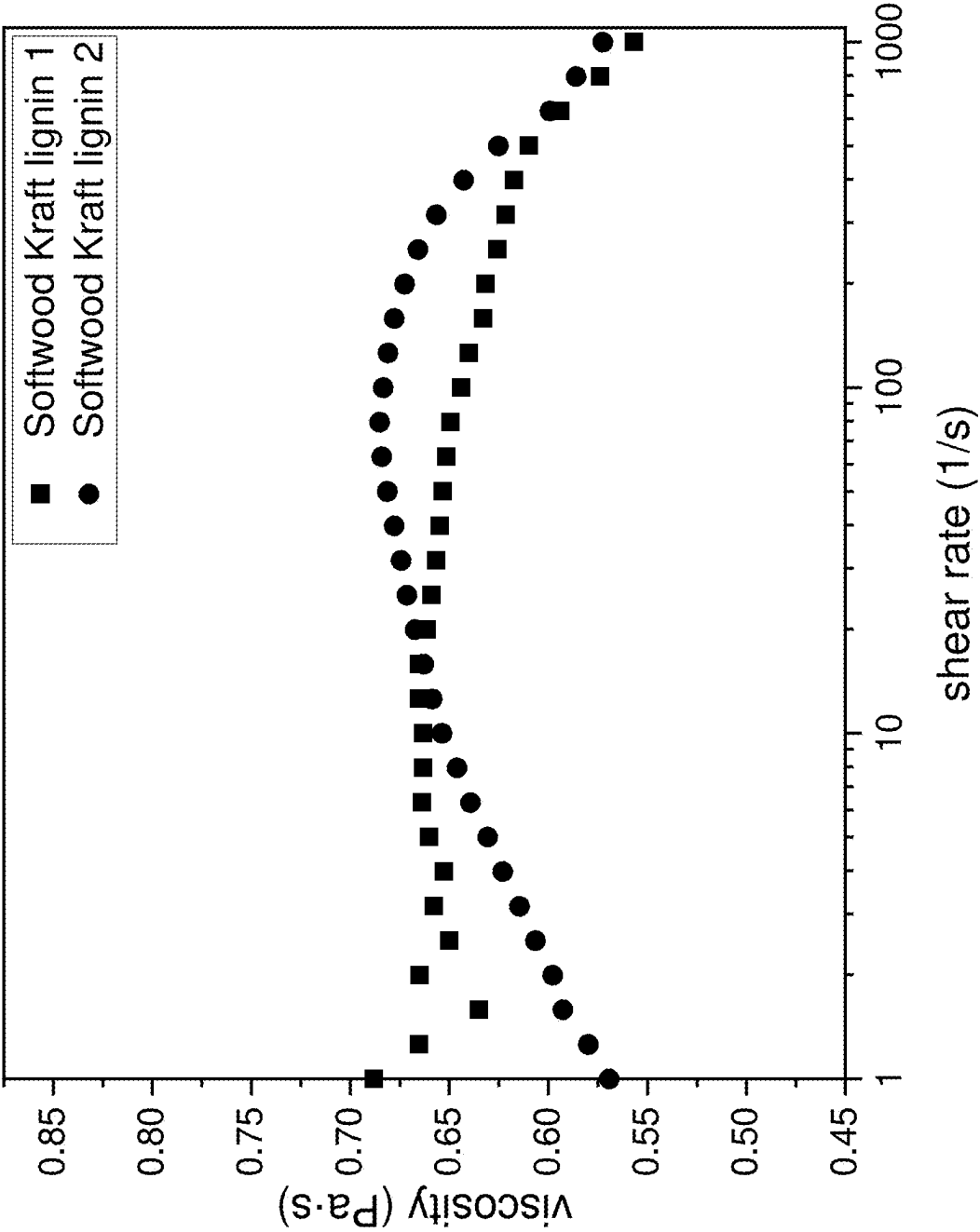


FIG. 8

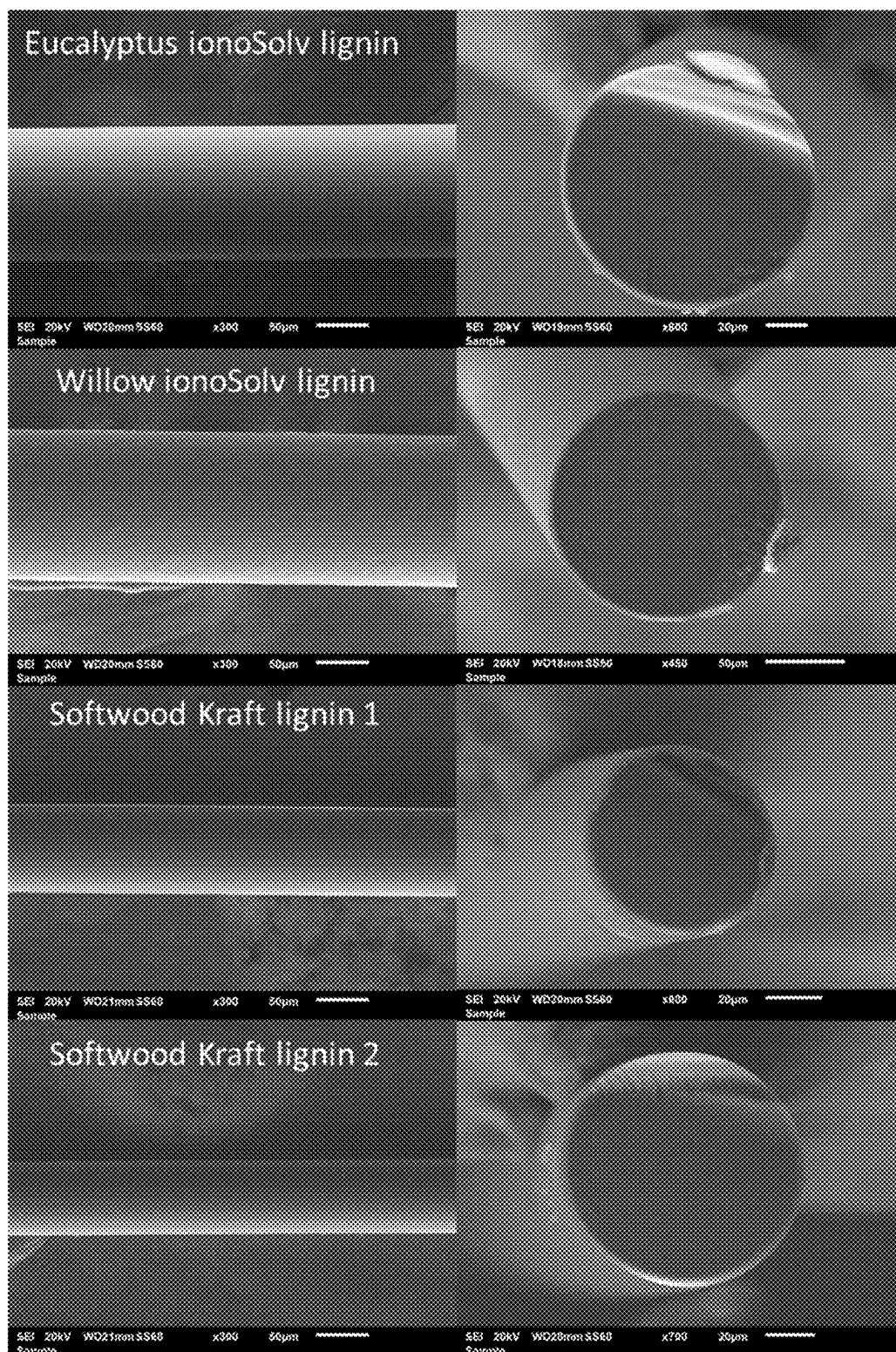


FIG. 9

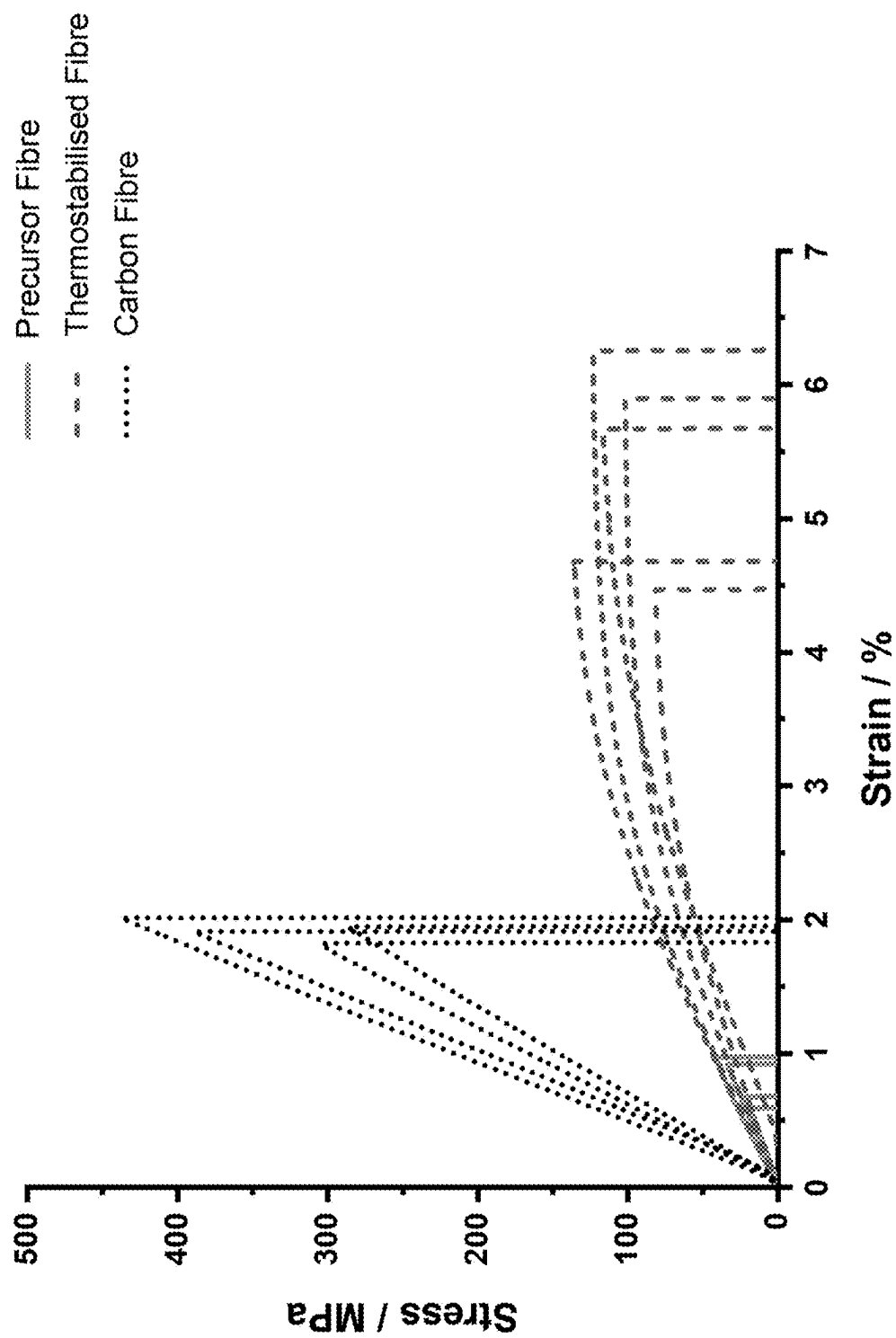


FIG. 10

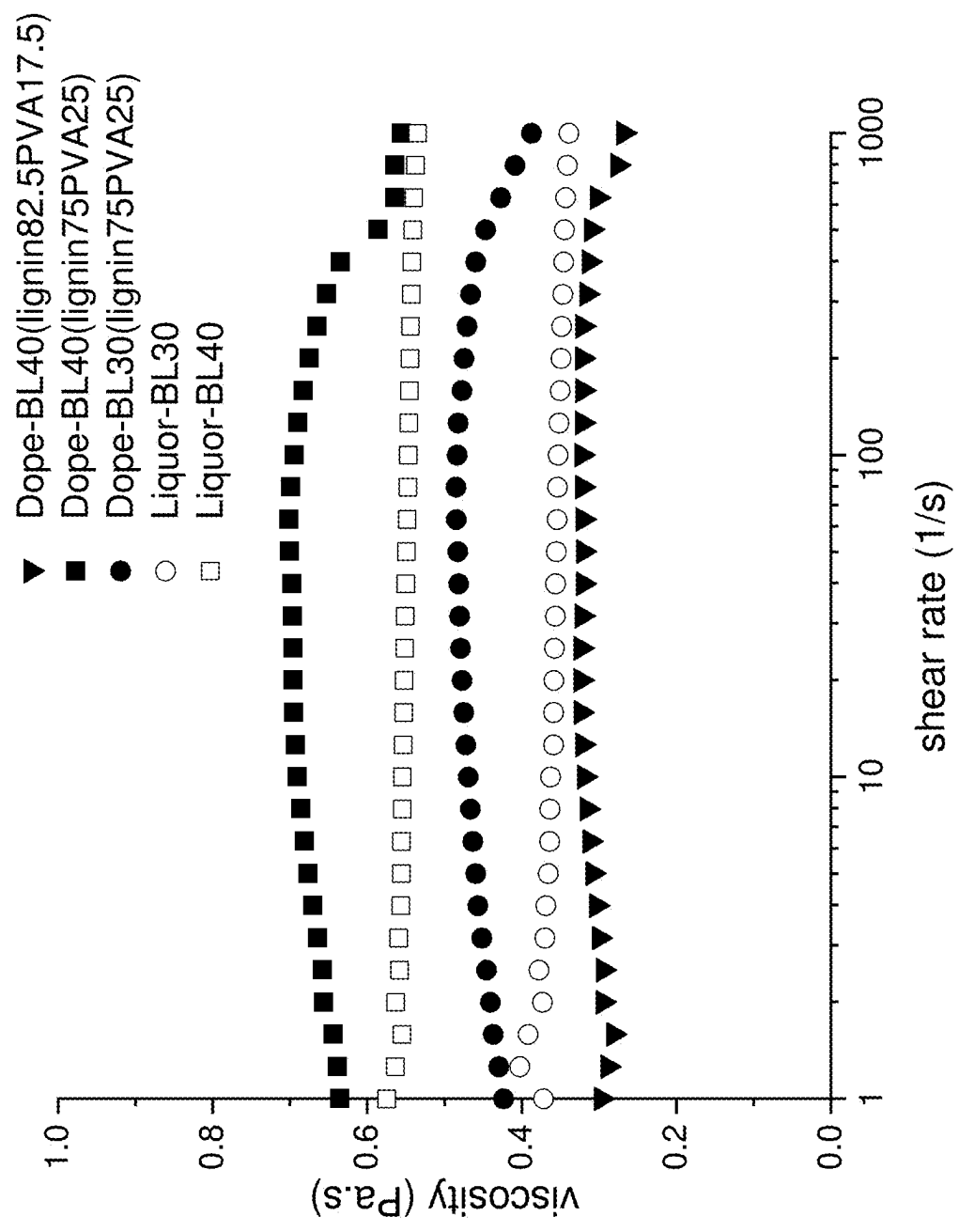


FIG. 11

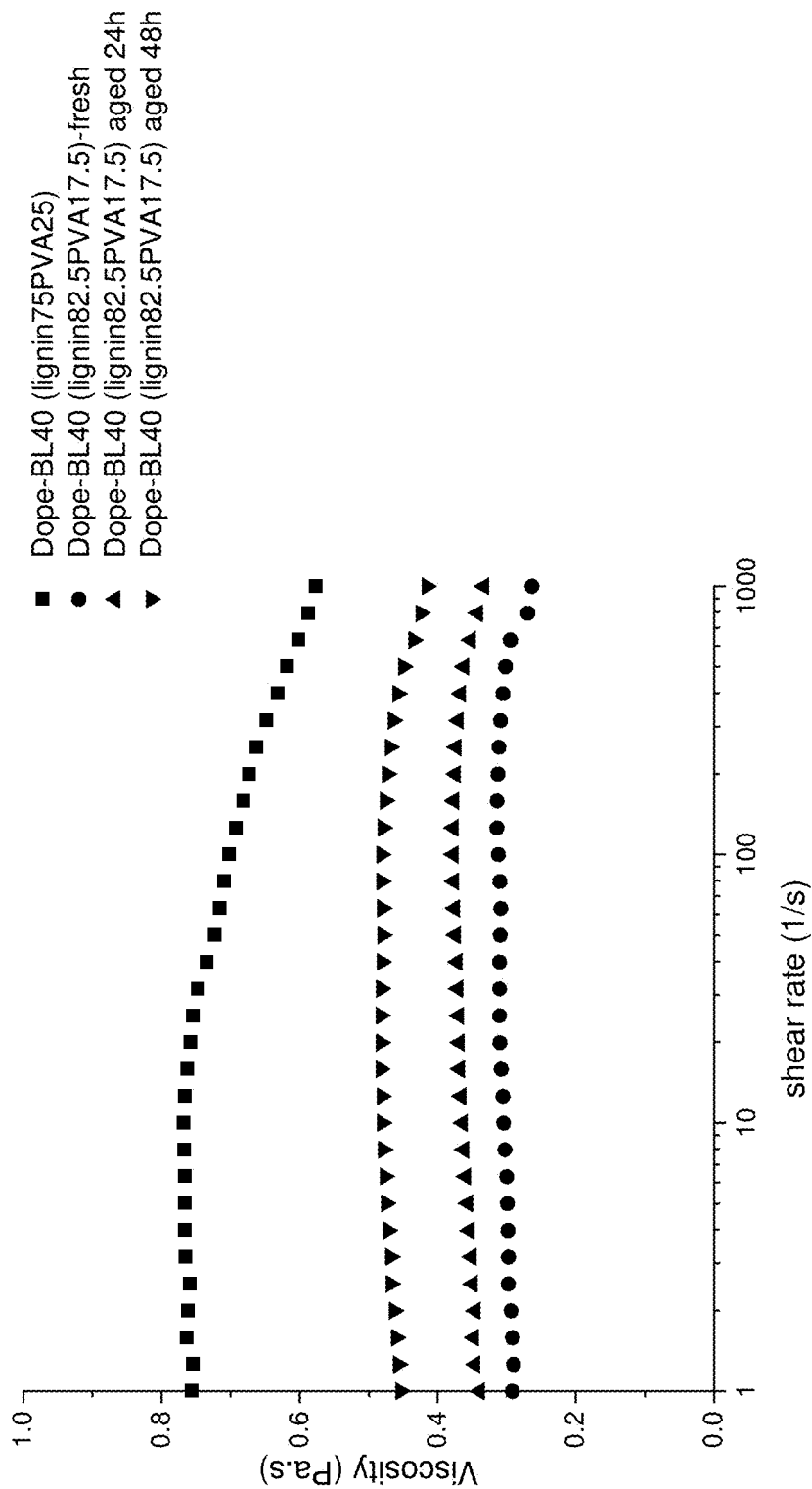


FIG. 12a

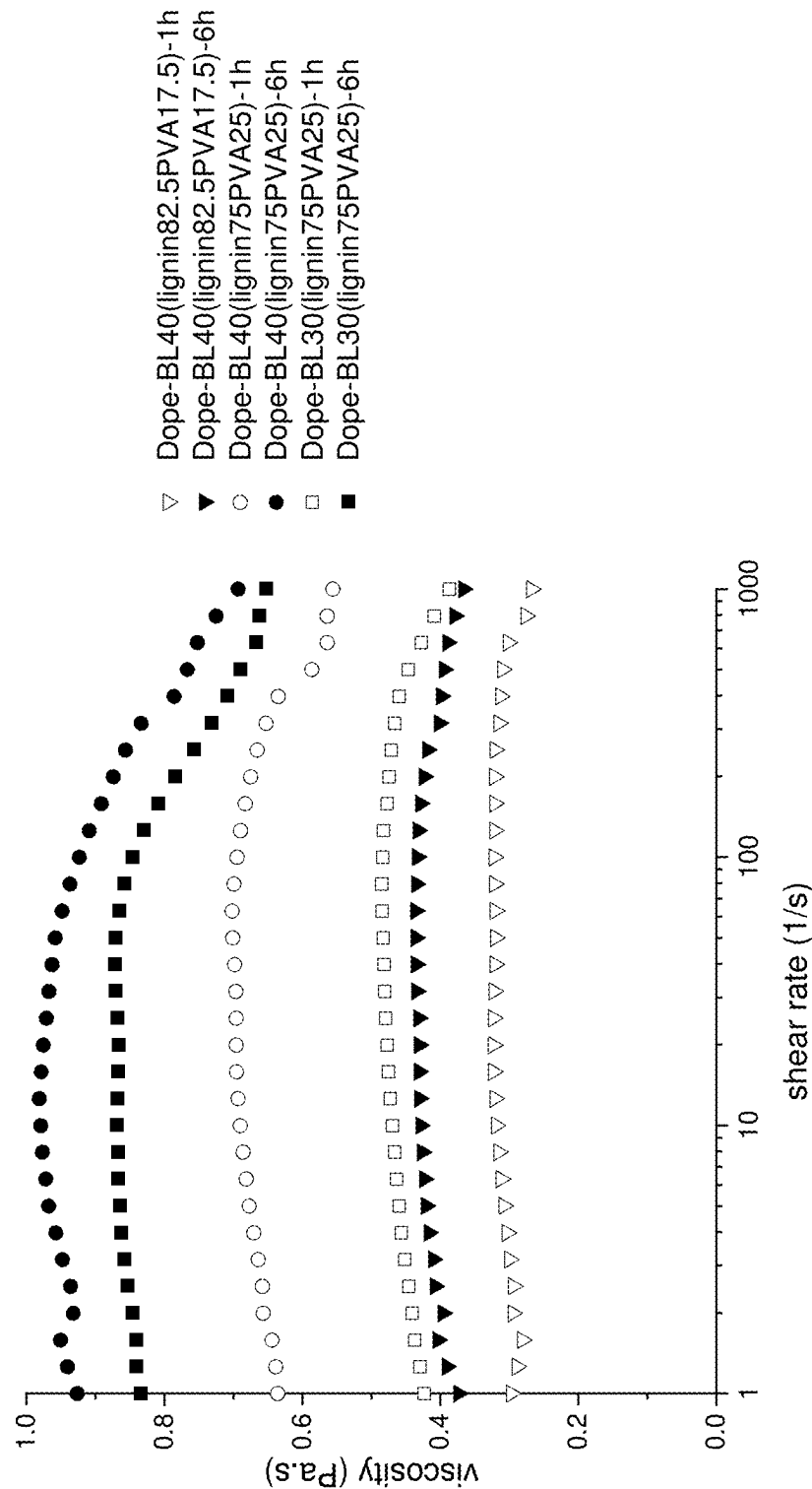


FIG. 12b

METHOD FOR MAKING FIBRES

FIELD OF THE INVENTION

[0001] This invention relates to a method for making fibres, the method comprising: providing a spinning dope comprising lignin and an additive polymer dissolved in a dope solvent, wherein the dope solvent comprises an ionic liquid and water; and extruding the spinning dope into a coagulation bath to obtain one or more fibres.

BACKGROUND TO THE INVENTION

[0002] Carbon fibres (CFs) are strong materials that can be used to produce carbon fibre reinforced composites, which are desirable lightweight construction materials. Carbon fibres are produced by pyrolysis of precursor fibres made from polyacrylonitrile (PAN) and mesophase petroleum pitch. However, the two major precursors are derived from petroleum and are, therefore, non-renewable. For PAN, the use of toxic spinning solvents such as DMF and generation of toxic by-products such as HCN during carbonisation raises additional environmental and health concerns. The high cost associated with precursor fabrication and the energy intensive high temperature processing also limits carbon fibre composite use to high-end markets and are an obstacle to fast market growth.

[0003] Lignin has the potential to be a lower cost and renewable alternative precursor, as it is a readily available biopolymer with a high carbon content. Lignin is attractive for its sustainable origin, low cost and relatively high fibre yield after carbonisation. Over 70 million tonnes of lignin are extracted each year during paper and pulp manufacture. Commercial lignin-based carbon fibres could support the economics of the developing renewable chemical industry by providing additional revenues to wood-processing biorefineries, which currently burn most of the lignin for generating heat and electricity rather than value added products.

[0004] Most research relating to the production of lignin fibres has focussed on melt spinning, at around 200° C., often with a co-polymer. Although the process is attractive as it avoids solvents, it is difficult to control the thermal behaviour of the lignin to obtain a suitable melt behaviour and oxidative stabilisation is slow to maintain fibre shape.

[0005] Wet (coagulation) spinning of pure unmodified lignin has not been demonstrated, likely due to its low average molar weight. Wet-spinning can be enabled by blending lignin with another fibre-forming polymer, such as cellulose. Solvents reported to date for wet-spinning include DMSO (Föllmer, M. et al, Wet-Spinning and Carbonization of Lignin-Polyvinyl Alcohol Precursor Fibers. *Advanced Sustainable Systems* 2019; Lu, C. et al, *ACS Sustainable Chemistry and Engineering* 2017, 5 (4), 2949-2959).

[0006] In addition, the ionic liquid (IL) 1-ethyl-3-methylimidazolium acetate, [Emim][OAc], has been used to form precursor lignin/cellulose fibres with pure water as a coagulant (Bengtsson, A. et al, *Holzforschung* 2018, 72 (12), 1007-1016; Vincent, S. et al., *ACS Sustainable Chemistry and Engineering* 2018, 6 (5), 5903-5910). The ionic liquid 1,5-diazabicyclo[4.3.0]non-5-enium acetate [DBNH][OAc] has also been used to produce carbon fibres derived from 50/50% Kraft lignin/cellulose precursor fibres (Ma, Y. et al. *ChemSusChem* 2015, 8 (23), 4030-4039). However, these

methods require expensive ILs that must be rigorously dried to dissolve cellulose, which represents a challenge for commercialisation.

[0007] Thus, there is a need for new lignin fibre spinning methods which enable the use of non-toxic, low-cost solvents that are water and moisture tolerant. It is also desirable to produce lignin fibres with higher carbon yield.

SUMMARY OF THE INVENTION

[0008] In a first aspect, provided herein is a method for making fibres, the method comprising: providing a spinning dope comprising a dope solvent, lignin dissolved in the dope solvent and an additive polymer dissolved in the dope solvent, wherein the dope solvent comprises an ionic liquid and water; and extruding the spinning dope into a coagulant to obtain one or more fibres.

[0009] In a second aspect, provided herein is a fibre obtainable by the method of the first aspect.

BRIEF DESCRIPTION OF FIGURES

[0010] Embodiments of the disclosure will now be described, by way of example only, and with reference to the drawings in which:

[0011] FIG. 1 shows the viscosity of dopes containing *Eucalyptus* ionoSolv lignin or Softwood Kraft lignin 1 prepared at 30° C. or 60° C.

[0012] FIG. 2 shows the diameters of ionoSolv *Eucalyptus* lignin fibres (lignin 75 wt %, PVA 25 wt %) spun from dopes prepared at different temperatures. Fibres were extruded at two different extrusion rates (ER).

[0013] FIG. 3a shows the tensile strengths and FIG. 3b shows the tensile moduli of *Eucalyptus* lignin precursor fibres (lignin 75 wt %, PVA 25 wt %). Fibres were extruded at two different extrusion rates (ER).

[0014] FIG. 4a shows the viscosity of dopes containing different amounts of lignin. FIG. 4b shows the viscosity of dopes aged for 0-6 h containing 75% lignin, measured at ca. 21.5° C.

[0015] FIG. 5 shows the viscosity change of *Eucalyptus* ionoSolv lignin dopes aged for different amounts of time.

[0016] FIGS. 6a-6d show the diameters and mechanical properties of precursor fibres tested at 15 mm gauge length containing different amount of *Eucalyptus* ionoSolv lignin, spun from dopes aged for different amounts of time.

[0017] FIG. 7 shows the steady shear viscosity as a function of shear rate of dopes containing different types of lignin.

[0018] FIG. 8 shows the viscosity of dopes containing two Kraft lignins.

[0019] FIG. 9 shows SEM images of precursor fibres derived from different types of lignin, coagulated in 1 M Na₂SO₄ (aq).

[0020] FIG. 10 shows stress strain curves for precursor lignin fibres (solid), thermostabilised fibres (dashed) and carbonised fibres (dotted) at 900° C. for 4 batches of each fibre type.

[0021] FIG. 11 shows the viscosity of the lignin containing liquor obtained with 30% biomass loading and the corresponding PVA containing dope; and viscosity of the liquor obtained with 40% biomass loading and two corresponding dopes with different lignin/PVA ratios.

[0022] FIGS. 12a and 12b show the effect of PVA aging at room temperature.

DETAILED DESCRIPTION

[0023] In a first aspect, provided herein is a method for making fibres, the method comprising: providing a spinning dope comprising a dope solvent, lignin dissolved in the dope solvent and an additive polymer dissolved in the dope solvent, wherein the dope solvent comprises an ionic liquid and water; and extruding the spinning dope into a coagulant to obtain one or more fibres.

[0024] The fibres produced by this method may be referred to as lignin fibres. The fibres comprise lignin and an additive polymer. The fibres may be used, for example, as precursor fibres for the production of carbon fibres or as a raw material for other fibre-based materials.

[0025] The spinning dope comprising both lignin and an additive polymer dissolved in a dope solvent. The spinning dope is preferably homogenous, without any undissolved components.

[0026] The dope solvent is a solvent within which lignin may be dissolved. Preferably, the dope solvent is a solvent within which cellulose has lower solubility than lignin. More preferably, the dope solvent does not dissolve cellulose.

[0027] The spinning dope preferably comprises a cellulose loading of no more than 10%, relative to the mass of spinning dope, excluding the mass of cellulose, lignin and additive polymer. Preferably, the cellulose loading is no more than 5%, no more than 2% cellulose or no more than 1%. The spinning dope is preferably substantially free of cellulose. Undissolved cellulose may be removed for example by filtration.

[0028] The dope solvent may have a water content of at least 5 wt %, for example 5-40 wt %, 10-40 wt % or 20-40 wt %. The water content of the dope solvent is calculated based on the mass of water present relative to the total mass of the dope solvent (i.e. w/w %). The presence of water reduces the overall solvent cost and minimises the energy requirements for solvent drying. The dope solvent may further comprise ethanol. The dope solvent may comprise 0-20 wt % ethanol, for example 1-20 wt % ethanol, preferably 5-15 wt % ethanol, calculated relative to the total mass of the dope solvent (i.e. w/w %).

[0029] Where the dope solvent comprises ethanol, the ionic liquid:ethanol mass ratio may be from 3:1 to 15:1, preferably from 5:1 to 12:1.

[0030] The spinning dope may have a total loading of lignin and additive polymer of 6-60 wt %, 6-50 wt %, 6-40 wt % or 6-30 wt %, for example 8-23 wt %, preferably 11-20 wt %, more preferably 10-20 wt %, even more preferably 15-20 wt %, calculated relative to the mass of mass of spinning dope, excluding the mass of lignin and additive polymer. This may also be referred to as the solid loading. A higher solid loading can reduce the production cost and increase dope viscosity, which may be beneficial for high-draw ratio spinning.

[0031] The weight ratio of lignin to polymer in the spinning dope may be at least 2:1 or at least 3:1, for example from 2:1 to 10:1, preferably from 3:1 to 9:1.

[0032] Lignin may be present in the spinning dope at a loading of at least 5 wt %, at least 7 wt % or at least 10 wt %, relative to the mass of the spinning dope, excluding the mass of lignin and additive polymer. Lignin may preferably be present at a loading of 5-50 wt %, 5-40 wt %, 5-20 wt %, 5-30 wt % or 5-15 wt %. Lignin may preferably be present at a loading of 7-30 wt %, 7-20 wt %, or 7-15 wt %, relative

to the mass of the spinning dope, excluding the mass of lignin and additive polymer. Lignin may preferably be present at a loading of 10-50 wt %, 10-40 wt %, 10-30 wt %, 10-20 wt % or 10-15 wt %, relative to the mass of the spinning dope, excluding the mass of lignin and additive polymer.

[0033] The lignin may be a hardwood lignin, softwood lignin, grass lignin or other lignin (e.g. a genetically modified lignin). The lignin may be a hardwood lignin. The lignin may be ionic liquid soluble lignin or kraft lignin, such as LignoBoost lignin.

[0034] The additive polymer may be present in the spinning dope at a loading of 1-10 wt %, relative to the mass of spinning dope, excluding the mass of lignin and additive polymer. The additive polymer may preferably be present at a loading of 1-5 wt %.

[0035] The spinning dope may have a water content of at least 5 wt %, for example 5-40 wt %, 10-40 wt % or 20-40 wt %, calculated based on the mass of water present relative to the mass of the spinning dope, excluding the mass of lignin and additive polymer.

[0036] The spinning dope may have a viscosity of 0.3-300,000, for example 0.3-100,000 or 0.3-2500 Pa·s, at zero shear (when measured at the spinning temperature). The zero-shear rate viscosity can be measured using an AR 2000ex rheometer with a cone-and-plate feature (2° cone angle, 20 mm plate diameter and 53 µm gap) at low shear rates from 3.00×10^{-6} to 30 s^{-1} at the spinning temperature. The spinning dope may have a viscosity of 0.3-12 Pa·s in the shear range of $1-1000 \text{ s}^{-1}$ and 0.3-6 Pa·s at a shear rate of ca. 183 s^{-1} , when measured using an AR 2000ex rheometer with a cone-and-plate feature (2° cone angle, 20 mm plate diameter and 53 µm gap) at shear rates from 1 to 1000 s^{-1} at the spinning temperature. The spinning temperature as referenced herein may refer to 25° C.

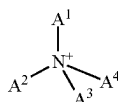
[0037] The additive polymer may be selected from poly(vinyl alcohol) (PVA), poly(vinyl acetate), poly(furfuryl alcohol), poly(acrylic acid), poly(ethylene oxide), poly(ethylene imine), poly(2-hydroxyethyl methacrylate), polyoxymethylene and mixtures thereof. Preferably, the polymer is PVA. PVA is able to hydrogen bond with the ionic liquid, such as [DMBA][HSO₄], to form a system-spanning PVA gel network. The network may comprise acid-catalysed crosslinked polymer chains. The PVA network acts as a higher molecular weight structure that lignin is able to hydrogen bond with to form a material that is no longer soluble in water. The gelling mechanism of PVA through hydrogen bonding in the presence of the ionic liquid, such as [DMBA][HSO₄], greatly influences the rheology of the dope solution, and therefore the ability to wet-spin continuous fibres.

[0038] The PVA may be partially hydrolysed. For example, the PVA may have a degree of hydrolysis (DH) of 72% or higher, such as 72-95% or 80-95%, preferably 85-90%, more preferably 86.7-88.7% hydrolysed. PVA may be formed by hydrolysis of poly(vinyl acetate) (PVAc). PVA may include partially hydrolysed PVA in which some PVAc monomeric units are not hydrolysed to PVA. The degree of hydrolysis is a mol % value of the OH groups relative to the amount of subunits (hydrolysed and acetylated) of PVA. Degree of hydrolysis may be measured by proton NMR spectroscopy.

[0039] The weight average molecular weight (Mw) of the additive polymer may be from 60 to 200 kDa, preferably 80 to 190 kDa.

[0040] The ionic liquid may be any ionic liquid as described herein. Preferably, the ionic liquid comprises a cation and an anion selected from C₁₋₂₀ alkyl sulfate ([AlkylSO₄]⁻), C₁₋₂₀ alkylsulfonate ([AlkylSO₃]⁻), hydrogen sulfate ([HSO₄]⁻), hydrogen sulfite ([HSO₃]⁻), dihydrogen phosphate ([H₂PO₄]⁻), hydrogen phosphate ([HPO₄]²⁻), chloride (Cl⁻), bromide (Br⁻), trifluoromethanesulfonate ([OTf]⁻), formate ([HCOO]⁻) and acetate ([MeCO₂]⁻). Preferably, the anion is selected from [HSO₄]⁻ and [HCOO]⁻.

[0041] The cation may be an aprotic cation or a protic cation, preferably a protic cation. The cation may contain a nitrogen-containing heterocyclic moiety or be a cation of Formula I



[0042] wherein A¹ to A⁴ are each independently selected from H, an aliphatic, C₃₋₆ carbocycle, C₆₋₁₀ aryl, alkylaryl, and heteroaryl.

[0043] The ionic liquid may be an [alkylammonium][HSO₄] or [alkylammonium][HCOO] ionic liquid.

[0044] The ionic liquid may be N,N-dimethylbutylammonium hydrogen sulfate ([DMBA][HSO₄]), 1-butylimidazolium hydrogen sulfate ([HBim][HSO₄]), triethylammonium hydrogen sulfate ([TEA][HSO₄]), N-methylbutylammonium hydrogen sulfate ([MBA][HSO₄]), 1-methylimidazolium formate ([HMim][HCOO]), N,N-dimethylbutylammonium formate ([DMBA][HCOO]), or a mixture thereof.

[0045] The ionic liquid may be [DMBA][HSO₄]. [DMBA][HSO₄] has shown particular aptitude in the extraction of lignin from lignocellulosic biomass fractionation while having a lower melting point (and hence viscosity) than other ionic liquids containing the hydrogen sulfate anion and a projected production cost of around \$1/kg (similar to [TEA][HSO₄]), making it cheaper than most ionic liquids and also DMSO. Moreover, ammonium-based hydrogen sulfate ILs may be recyclable.

[0046] In the method described herein, fibres are formed using wet (coagulation) spinning. This involves extruding a spinning dope into a coagulant to form fibres. The coagulant may be contained in a coagulation bath. The coagulant may comprise water, preferably deionised (DI) water. The coagulant may consist essentially of water. As an alternative, the coagulant may comprise water and an ionic liquid. The ionic liquid may be any ionic liquid as described herein and, preferably, the same ionic liquid as present in the dope solvent. The ionic liquid may be present within the coagulant at no more than 60 wt %, no more than 30 wt %, or no more than 15 wt %, preferably 1-60 wt %, 1-30 wt % or 1-15 wt %, more preferably 5-10 wt % (based on total mass of coagulant). In a further alternative, the coagulant comprises water and sodium sulfate (Na₂SO₄). For example, the coagulation bath may comprise aqueous sodium sulfate at a concentration of 0.5-1.5 M, preferably about 1 M.

[0047] The method described herein may utilise a coagulation time of at least 30 seconds, at least 45 seconds, at least 60 seconds, or at least 90 seconds, preferably at least 45

seconds. The coagulation time represents the residence time of the spinning dope in the coagulant to form fibres. After the coagulation time, fibres may be handled and/or collected from the coagulant.

[0048] The spinning dope may be prepared by a process comprising: a) providing the ionic liquid, optionally in a mixture with water, adding an aqueous solution of the additive polymer and, optionally, adding water to reach the desired water content to provide a solution of additive polymer in the dope solvent; and b) adding lignin to the solution formed in step a) to dissolve the lignin and provide the spinning dope.

[0049] Dissolution of the lignin may be performed at a temperature of 10° C. to 200° C., preferably 10° C. to 100° C., more preferably 20° C. to 100° C., 20° C. to 60° C., or 20° C. to 30° C.

[0050] The method may further comprise aging the spinning dope for at least 2 minutes, at least 5 minutes, at least 30 minutes, preferably 6-48 hours, more preferably 6-24 hours, prior to the extruding. Aging may be carried out at room temperature.

[0051] The spinning dope may be heated during aging, for example, at a temperature of 20° C. to 90° C., 30° C. to 90° C. or 30° C. to 60° C., or a temperature of at least 30° C., at least 60° C., e.g. up to 150° C. The spinning dope may be heated prior to extruding (during aging) for at least 2 minutes, at least 5 minutes, at least 30 minutes or 30 minutes to 10 hours, for example 1 to 6 hours. These heating periods may correspond to the duration of aging.

[0052] The method may further comprise mixing the spinning dope during aging, preferably for at least 2 minutes, at least 5 minutes, at least 30 minutes or 30 minutes to 10 hours, for example 1 to 6 hours.

[0053] Mixing, heating and ageing of the spinning dope prior to extruding can be used to improve spinnability of the spinning dope.

[0054] The fibres may be extruded using any suitable wet-spinning process, for example using a continuous spinning line. The fibres may be extruded into a rotational bath. A rotation bath is lab based approach of spinning fibre from dopes ensuring constant draw. However, unlike the standard fibre spinning process using a continuous spinning line where different degree of drawing could be achieved by changing ratio of the winding rate to injection rate (draw ratio), using a rotational bath limits the drawing as the maximum drawing provided is dependent on the viscous force of the coagulant.

[0055] The size of needle/1-hole spinneret used to spin lignin fibres with a spinning line may for example be 300 µm-1 mm (internal diameter) when extensional drawing can be achieved with lab-based equipment. A thinner needle (such as a 27 G needle having an internal diameter of 210 µm) may be used to help align the polymer chains in the fibre and reduce the fibre diameter when the drawing is limited. Even thinner orifices (e.g. needles) may be used, for example, in a continuous spinning line.

[0056] The spinning dope may be prepared by a process comprising: a) contacting a lignocellulosic biomass comprising lignin and cellulose with a composition comprising the ionic liquid and water to dissolve the lignin and produce a cellulose pulp; b) separating the cellulose pulp to obtain a liquor comprising the ionic liquid, water and lignin; and c) combining the liquor with the additive polymer to obtain the spinning dope. In step a) lignin is dissolved and cellulose

remains undissolved, to produce a cellulose pulp. The ionic liquid is the same ionic liquid as present within the dope solvent. Step c) of combining the liquor with the additive polymer may comprise combining the liquor with an aqueous solution of the additive polymer. The liquor produced in step b) may be concentrated prior to step c). The composition comprising the ionic liquid and water (also referred to as the ionic liquid/water composition) referenced in step (a) may comprise a 2-40 wt % water content, a 5-40 wt % water content, a 5-30 wt % water content, or a 10-30 wt % water content. The ionic liquid/water composition may consist essentially of ionic liquid and water. The water content referenced in step (a) is calculated based on the mass of water present relative to the total mass of the ionic liquid/water composition. The biomass loading in step a) may be, for example, 10-50 wt % or 20-50 wt %, such as 30-40 wt %, relative to the mass of the ionic liquid/water composition. Steps (a)-(c) make lignin extraction, formation of the spinning dope, and fibre formation possible without requiring separate steps of isolating and/or drying lignin. This may be referred to as an integrated spinning process. This approach avoids the need to recycle the ionic liquid after the extraction of lignin. As a result, the ionic liquid needs to be recycled only once, after fibre spinning. By avoiding lignin precipitation and drying of the precipitated lignin and the ionic liquid after the lignin precipitation step, this approach has the potential to lower the cost of precursor fibre production. The complete coagulation of lignin and the additive polymer into the fibre allows any residual hemicellulose components to diffuse out of the lignin co-polymer matrix, allowing hemicellulose valorisation as acetic acid and furfural.

[0057] The lignocellulosic biomass contacted with the composition in step a) may be heated to a temperature of at least 70° C., preferably 100-180° C., more preferably 120-170° C. For example, the lignocellulosic biomass contacted with the composition may be heated to 120-150° C. Heating may be carried out for 1 minute to 22 hours, 10 minutes to 20 hours, 10 minutes to 10 hours, 15 minutes to 8 hours or 30 minutes to 8 hours.

[0058] The biomass may be contacted with the composition and subjected to mechanical treatment, such as stirring or vortexing, to aid dissolution of the lignin and production of a cellulose pulp. Mechanical treatment may be carried out prior to heating. Ethanol may be added to the mixture resulting from step a) prior to separation of the cellulose pulp. Separation of the cellulose pulp may be carried out using filtration, for example vacuum filtration. The biomass may undergo mechanical processing before being contacted with the composition.

[0059] The integrated spinning process results in dissolution of lignin from lignocellulosic biomass in the dope solvent, but avoids dissolution of cellulose. Other components of the lignocellulosic biomass, such as hemicellulose, may dissolve.

[0060] The spinning dope may comprise additional solutes, wherein the additional solutes may be lignocellulosic biomass components such as hemicellulose, or hemicellulose degradation products, such as furfural.

[0061] The method may further comprise washing the one or more fibres after extrusion. The washing may be carried out with water.

[0062] The method may further comprise drying the one or more fibres, preferably under mechanical tension. This step may be carried out after the washing step.

[0063] The method may further comprise heating the one or more fibres in air at 150-300° C. This step may be performed to thermally stabilise the one or more fibres.

[0064] The thermally stabilised fibres may be used to form a fabric. Accordingly, the method may further comprise weaving the thermally stabilised fibres to form a fabric.

[0065] The method may further comprise carbonising the one or more fibres to obtain carbon fibres. The carbonising may comprise heating the one or more fibres to 800-3000° C., preferably 1200-1800° C., under an inert atmosphere. For example, the carbonising may be performed under nitrogen or argon, preferably nitrogen. Carbonising may be carried out with the fibres under tension. Carbonising may be carried out after thermal stabilisation.

[0066] In a second aspect, provided herein is a fibre obtainable by the method of the first aspect.

Ionic Liquids

[0067] The ionic liquid referenced herein may, for example, be an ionic liquid as described in WO2012080702, WO2014140643 or WO2017085516, which are incorporated herein by reference.

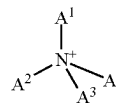
[0068] As used herein “ionic liquid” refers to an ionized species (i.e. cations and anions). Ionic liquids typically have a melting point below about 100° C. Any of the anions listed below can be used in combination with any of the cations listed below, to produce an ionic liquid for use in the invention.

[0069] The ionic liquid may contain one of the listed anions, or a mixture thereof.

[0070] The anion may be selected from C₁₋₂₀ alkyl sulfate ([AlkylSO₄]⁻), C₁₋₂₀ alkylsulfonate ([AlkylSO₃]⁻), hydrogen sulfate ([HSO₄]⁻), hydrogen sulfite ([HSO₃]⁻), dihydrogen phosphate ([H₂PO₄]⁻), hydrogen phosphate ([HPO₄]²⁻), chloride (Cl⁻), bromide (Br⁻), trifluoromethanesulfonate ([OTf]⁻), formate ([HCOO]⁻) and acetate ([MeCO₂]⁻). For example, the anion may be selected from [MeSO₄]⁻, [HSO₄]⁻, [MeSO₃]⁻, Cl⁻, [HCOO]⁻ and [MeCO₂]⁻, such as from chloride Cl⁻ and hydrogen sulfate [HSO₄]⁻, or from [MeSO₄]⁻, [HSO₄]⁻, [MeSO₃]⁻, and [MeCO₂]⁻. Preferably the anion is selected from [HSO₄]⁻ and [HCOO]⁻. The ionic liquid may contain one of the listed anions, or a mixture thereof. The ionic liquid may contain any one of the cations identified herein, or a mixture thereof.

[0071] The cation is preferably a protic cation ion i.e. The cation is capable of donating a proton (H⁺).

[0072] The cation may be an ammonium or phosphonium derivative. These cations have the general formula

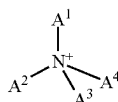


[0073] wherein

[0074] X is N or P; and

[0075] A¹ to A⁴ are each independently selected from H, an aliphatic, C₃₋₆ carbocycle, C₆₋₁₀ aryl, alkylaryl, and heteroaryl. The aliphatic may optionally be substituted with one or more —OH.

[0076] In an embodiment, the cation is an ammonium ion, a derivative thereof or a mixture thereof. The cation may be of the formula



[0077] wherein

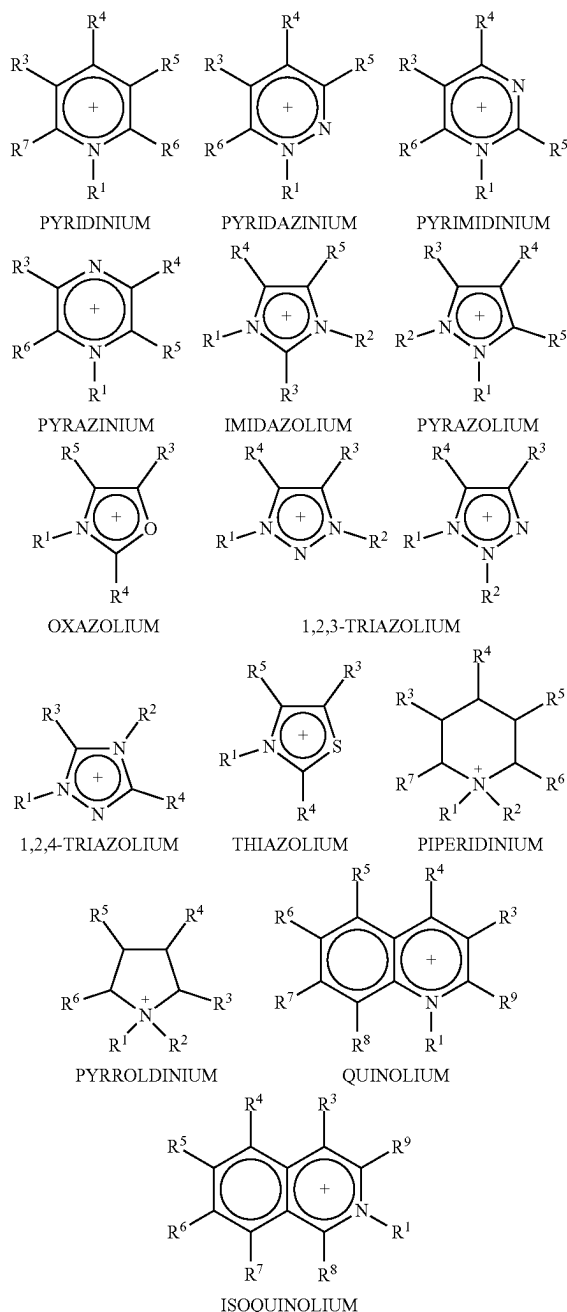
[0078] A^1 to A^4 are each independently selected from H, an aliphatic, C_{3-6} carbocycle, C_{6-10} aryl, alkylaryl, and heteroaryl, wherein aliphatic may optionally be substituted with one or more-OH. Preferably at least one of A^1 to A^4 is H. Preferably A^1 to A^4 are each independently selected from H, and an aliphatic. In one embodiment one of A^1 to A^4 is H, and the remaining three are each independently an aliphatic. Alternatively, two of A^1 to A^4 are each H and the remaining two are each independently an aliphatic. Alternatively, one of A^1 to A^4 is an aliphatic, and the remaining three are all H. Preferably the cation is not ammonium (NH_4^+) i.e. at least one of A^1 to A^4 is not H. Aliphatic may be alkyl, optionally substituted with one or more-OH, preferably C_{1-6} alkyl. In some embodiments, the aliphatic is unsubstituted.

[0079] In an embodiment, the cation is an alkylammonium or a mixture thereof (i.e. a cation of the formula above wherein A^1 to A^4 are each independently selected from H or alkyl, wherein at least one of A^1 to A^4 is not H). Preferably this is a protic alkylammonium, although aprotic alkylammoniums may also be used. Optionally one or more of the alkyl groups may be substituted with —OH to form an alkanolammonium, which can also be referred to as an alcoholammonium. For example, the cation may be choline. As used herein an “alkylammonium” includes trialkylammoniums, dialkylammoniums, monoalkylammoniums, and alcoholammoniums including trialcoholammoniums, dialcoholammoniums and monoalcoholammonium. Trialkylammoniums include trimethylammonium, triethylammonium, and triethanolammonium. Examples of dialkylammoniums include diethylammonium, diisopropylammonium, and diethanolammonium. Monoalkylammoniums include methylammonium, ethylammonium, and monoethanolammonium. The ionic liquid may preferably be an [alkylammonium][HSO_4] or an [alkylammonium][$HCOO$] ionic liquid.

[0080] In an embodiment, the alkylammonium cation is selected from triethylammonium, diethylammonium dimethylethylammonium, diethylmethylammonium, dimethylbutylammonium, diethanolammonium and choline. The alkylammonium cation may be selected from triethylammonium, diethylammonium dimethylethylammonium, diethylmethylammonium, and dimethylbutylammonium, diethanolammonium. In another embodiment, the alkylammonium cation is selected from dimethylbutylammonium, triethylammonium and methylbutylammonium.

[0081] The cation can also contain a nitrogen-containing heterocyclic moiety which, as used herein, refers to mono- or bicyclic ring systems which include one nitrogen atom and optionally one or more further heteroatoms selected from N, S and O. The ring systems contain 5-9 members, preferably 5 or 6 members for monocyclic groups, and 9 or 10 members for bicyclic groups. The rings can be aromatic,

partially saturated or saturated and thus, include both a “heteroalicyclic” group, which means a non-aromatic heterocycle and a “heteroaryl” group, which means an aromatic heterocycle. The cation may be selected from



[0082] wherein R^1 and R^2 are independently a C_{1-6} alkyl or a C_{1-6} alkoxyalkyl group, and R^3 , R^4 , R^5 , R^6 , R^7 , R^8 and R^9 , when present are independently H, a C_{1-6} alkyl, C_{1-6} alkoxyalkyl group, or C_{2-6} alkoxy group. Preferably R^1 and R^2 are C_{1-4} alkyl, with one being methyl and R^3 - R^9 , (R^3 , R^4 , R^5 , R^6 , R^7 , R^8 and R^9), when present, are H. In an embodiment, the cation ring is imidazolium or pyridinium.

[0083] In an embodiment, the cation may be an imidazolium based cation or a mixture thereof, in particular protic imidazolium based cations. In an embodiment, the imidazolium based cation may be selected from 1-butyl-3-methylimidazolium [BMim]⁺, 1-ethyl-3-methylimidazolium [EMim]⁺, 1-methylimidazolium [HMim]⁺, 1-butylimidazolium [HBim]⁺ and mixtures thereof. For example, the imidazolium based cation may be selected from 1-butyl-3-methylimidazolium [BMim]⁺, 1-methylimidazolium [HMim]⁺, 1-butylimidazolium [HBim]⁺ and mixtures thereof, such as 1-methylimidazolium [HMim]⁺, 1-butylimidazolium [HBim]⁺ and mixtures thereof. In an embodiment, the imidazolium based cation is selected from 1-butyl-3-methylimidazolium [BMim]⁺, 1-butylimidazolium [HBim]⁺ and mixtures thereof.

[0084] In some embodiments, the cations include protic alkylammonium, protic methylimidazolium, protic pyridinium, aprotic tetraalkylammonium and aprotic dialkylimidazolium ions.

[0085] The ionic liquid may preferably be N,N-dimethylbutylammonium hydrogen sulfate ([DMBA][HSO₄]), 1-butylimidazolium hydrogen sulfate ([HBim][HSO₄]), triethylammonium hydrogen sulfate ([TEA][HSO₄]), methylbutylammonium hydrogen sulfate ([MBA][HSO₄]), 1-methylimidazolium formate ([HMim][HCOO]), N,N-dimethylbutylammonium formate ([DMBA][HCOO]), or a mixture thereof.

[0086] In an embodiment, the ionic liquid is not 1-ethyl-3-methylimidazolium acetate [EMim][OAc].

[0087] In another embodiment, the ionic liquid is selected from triethylammonium hydrogen sulfate [TEA][HSO₄], N,N-dimethylbutylammonium hydrogen sulfate [DMBA][HSO₄], diethylammonium hydrogen sulfate [DEA][HSO₄], N,N-dimethylethylammonium hydrogen sulfate ([DMEA][HSO₄]), Diethanolammonium chloride [DEtOHA]Cl, 1-methylimidazolium hydrogen chloride [HMim]Cl, 1-ethyl-3-methylimidazolium chloride [EMim]Cl, and 1-ethyl-3-methylimidazolium trifluoromethanesulfonate [EMim][OTf].

[0088] In another embodiment, the ionic liquid is selected from 1-butyl-3-methylimidazolium methyl sulfate [BMim][MeSO₄], 1-butyl-3-methylimidazolium hydrogen sulfate [BMim][HSO₄], 1-butyl-3-methylimidazolium methane-sulfonate [BMim][MeSO₃], 1-butylimidazolium hydrogen sulfate [HBim][HSO₄], and 1-ethyl-3-methylimidazolium acetate [EMim][MeCO₂].

[0089] Preferred ionic liquids are [alkylammonium][HSO₄] ionic liquids, for example N,N-dimethylbutylammonium hydrogen sulfate ([DMBA][HSO₄]), triethylammonium hydrogen sulfate ([TEA][HSO₄]), N-methylbutylammonium hydrogen sulfate ([MBA][HSO₄]), diethylammonium hydrogen sulfate [DEA][HSO₄], N,N-dimethylethylammonium hydrogen sulfate ([DMEA][HSO₄]), and ethylammonium hydrogen sulfate [EA][HSO₄].

[0090] The dope solvent referenced herein comprises ionic liquid and water. For use in the method disclosed herein, an ionic liquid is provided such that the dope solvent is capable of dissolving lignin. Preferably, cellulose has lower solubility than lignin in the dope solvent. More preferably, the dope solvent does not dissolve cellulose.

[0091] The ionic liquid, optionally in a mixture with water, may be used in the treatment of lignocellulosic biomass, for example to separate lignin and cellulose in preparation of the

spinning dope in an integrated spinning process. The ionic liquid may dissolve the lignin within the biomass but not the cellulose, so that the treatment yields a cellulose pulp and lignin solution. Thus, the majority of the cellulose remains solid, for example at least 70%, preferably at least 80% (wt % relative to oven dried weight of biomass). The cellulose pulp can be easily removed from the lignin solution mechanically, for example by filtration. Other components such as hemicellulose may also dissolve in the ionic liquid. For example, at least 70%, preferably at least 80%, of the hemicellulose in the lignocellulosic biomass may be dissolved in the ionic liquid.

[0092] When the ionic liquid is present in a mixture with water, this may be expressed as [ionic liquid]_x %/water_y %, wherein the percentage is the mass of the component relative to the total mass of the mixture (i.e. w/w %). For example, [DMBA][HSO₄]_{60%}/water_{40%} refers to a mixture of [DMBA][HSO₄] and water, wherein the [DMBA][HSO₄] is present at 60% (w/w) (60 wt %) and the water is present at 40% (w/w) (40 wt %).

[0093] Ionic liquids can be prepared by methods known to the person skilled in the art or obtained commercially. For example, protic ammonium-based ILs can be made from a simple alkylamine, such as triethylamine, and sulfuric acid in a one-step synthesis, for example as described in George et al., (2015) "Design of low-cost ionic liquids for lignocellulosic biomass treatment" *Green Chemistry* 17:1728-173.

[0094] Usually in an ionic liquid, the cation and anion are present in equimolar amounts. However, the ionic liquid may comprise excess base, preferably protonated base. 'Base' as used herein refers to the base from which the cation is derived e.g. amine/imidazole. The ionic liquid may comprise 10% molar excess base, for example, 4-8%, 5-7.5% excess base. The ionic liquid may comprise 2%, 3%, 4%, 5%, 6%, 7%, 8% 9% or 10% molar excess base.

[0095] In another embodiment, the dope solvent further comprises 0.01-20% molar excess acid, preferably 1-5% molar excess acid, as a percentage of the IL. The acid can be selected from any known strong acid such as hydrochloric acid, sulfuric acid, nitric acid, phosphoric acid hydroiodic acid, perchloric acid and hydrobromic acid. Preferably the acid is sulfuric or, hydrochloric or phosphoric acid. More preferably, the acid is the same acid as used to synthesise a protic IL.

[0096] It should be appreciated that the features described throughout this disclosure may be present in any combination mutatis mutandis. For example, wherever discussion of the components or component loadings of the spinning dope is provided, these components can be present in any combination and the loadings may be present in any combination.

[0097] All publications, published patent documents, and patent applications cited herein are hereby incorporated by reference to the same extent as though each individual publication, published patent document, or patent application was specifically and individually indicated as being incorporated by reference.

Definitions

[0098] In order for the present invention to be more readily understood, certain terms are first defined below. Additional definitions for the following terms and other terms are set forth throughout the specification.

[0099] As used herein, singular forms “a,” “an” and “the” also include plural forms unless the context clearly dictates otherwise. Use of the singular includes the plural unless specifically stated otherwise. The terms “comprising,” “containing” and “including” as well as other forms (e.g., “include,” “comprise” and “contain”) are not limiting. As used herein, wherever “comprising” is referenced, this may also refer to “consisting essentially of” and “consisting of”. For example, a dope solvent comprising ionic liquid, water and optionally ethanol, may also be a dope solvent consisting essentially of ionic liquid, water and optionally ethanol.

[0100] A “dope solvent” as referenced herein is a solvent mixture within which lignin may be dissolved. The dope solvent comprises an ionic liquid and water. It may comprise additional solvents, such as ethanol. All solvent present within the spinning dope may constitute the dope solvent (i.e. the spinning dope does not contain any solvent other than the dope solvent). The dope solvent may consist essentially of ionic liquid, water and optionally ethanol. Preferably, the dope solvent does not dissolve cellulose.

[0101] Various components are described as being present in the spinning dope at a percentage loading. The loading is a mass loading, that may be referred to as a % loading or a wt % loading. In general terms, the loading is described as being relative to the mass of the spinning dope, excluding the mass of lignin and additive polymer. The loading calculation for a spinning dope component (e.g. lignin or additive polymer) is:

$$\frac{(\text{mass of component}/\text{mass of spinning dope excluding mass of lignin and additive polymer}) \times 100}{100}$$

[0102] In embodiments where the spinning dope consists essentially of dope solvent (water, ionic liquid and optionally ethanol), lignin and additive polymer, the loading calculation is: (mass of component/dope solvent)×100

[0103] As described herein, in some embodiments, the spinning dope may comprise additional solutes. In such embodiments, the loading calculation is:

$$\frac{(\text{mass of component}/\text{dope solvent} + \text{additional solutes}) \times 100}{100}$$

[0104] As used herein, a “coagulant” is a liquid into which a spinning dope can be extruded. Extrusion of the spinning dope into the coagulant results in fibre formation.

[0105] An “additive polymer” as used herein is a polymer included within the spinning dope to aid in formation of lignin fibres. The additive polymer is not a polymer derived from lignocellulosic biomass. Accordingly, the additive polymer is not lignin, cellulose or hemicellulose, or a cellulose or hemicellulose derivative or degradation product (e.g. a hemicellulose sugar).

[0106] As used herein the term “solid loading” refers to the weight ratio of solid to solvent, expressed as a percentage. Thus, the solid loading of a spinning dope comprising lignin and an additive polymer dissolved in a dope solvent may be expressed as follows:

$$\frac{m_{\text{lignin}} + m_{\text{additive polymer}}}{m_{\text{dope solvent}}} \times 100\%$$

[0107] wherein m_{lignin} is the mass of the lignin, $m_{\text{additive polymer}}$ is the mass of the additive polymer and $m_{\text{dope solvent}}$ is the mass of the dope solvent.

[0108] Room temperature as referenced herein may refer to 25° C.

[0109] As used herein the term poly(vinyl alcohol) (PVA) includes fully hydrolysed PVA and partially hydrolysed PVA. The degree of hydrolysis is expressed as mol %. In partially hydrolysed PVA, some PVAc monomeric units are not hydrolysed to PVA. The degree of hydrolysis is a mol % value of the OH groups relative to the amount of subunits (hydrolysed and acetylated) of PVA. Degree of hydrolysis may be measured by proton NMR spectroscopy.

[0110] A polymer’s molecular weight may be specified as a weight average molecular weight (Mw) or a number average molecular weight (Mn). Molecular weights may be determined, for example, by gel permeation chromatography.

[0111] As used herein the term “lignocellulosic biomass” refers to living or dead biological material and can comprise any cellulosic or lignocellulosic material such as cellulose, optionally further comprising hemicellulose, lignin, starch, oligosaccharides and/or monosaccharides, biopolymers, natural derivatives of biopolymers, their mixtures, and breakdown products. It can also comprise additional components, such as protein and/or lipid. The biomass can be derived from a single source, or it can comprise a mixture derived from more than one source. Some specific examples of biomass include, but are not limited to, bioenergy crops, agricultural residues, municipal solid waste, industrial solid waste, sludge from paper manufacture, yard waste, wood and forestry waste. Additional examples of biomass include, but are not limited to, corn grain, corn cobs, crop residues such as corn husks, corn stover, grasses including *Miscanthus X giganteus*, wheat, wheat straw, hay, rice straw, switchgrass, waste paper, sugar cane bagasse, sorghum, soy, components obtained from milling of grains, trees (e.g. pine), branches, roots, leaves, wood chips, wood pulp, sawdust, shrubs and bushes, vegetables, fruits, flowers, animal manure, multi-component feed, and crustacean biomass (i.e., chitinous biomass). It may be preferable to treat the biomass before use in the method of the invention. For example the biomass could be mechanically treated e.g. milling or shredding.

[0112] As referenced herein, “aging” refers to a period of time after the spinning dope has been prepared before the dope is extruded. During aging, the dope may remain at room temperature or may be heated. During aging, the dope may undergo mixing. Aging may occur for at least 30 minutes, preferably up to 72 or 48 hours.

[0113] The term “aliphatic” as used herein refers to a straight or branched chain hydrocarbon which is completely saturated or contains one or more units of unsaturation. Thus, aliphatic may be alkyl, alkenyl or alkynyl, preferably having 1 to 12 carbon atoms, preferably up to 6 carbon atoms or more preferably up to 4 carbon atoms. The aliphatic can have 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 carbon atoms.

[0114] The term “alkyl” as used herein, is typically a linear or branched alkyl group or moiety containing from 1 to 20 carbon atoms, such as 11, 12, 13, 14, 15, 16, 17, 18, or 19 carbon atoms. Preferably the alkyl group or moiety contains 1-10 carbon atoms, i.e., 2, 3, 4, 5, 6, 7, 8, 9, or 10 carbon atoms such as a C₁₋₄ alkyl or a C₁₋₆ alkyl group or moiety, for example methyl, ethyl, n-propyl, i-propyl, n-butyl, i-butyl and t-butyl, n-pentyl, methylbutyl, dimethylpropyl, n-hexyl, 2-methylpentyl, 3-methylpentyl, 2,3-dimethylbutyl, and 2,2-dimethylbutyl.

[0115] The term “carbocycle” as used herein refers to a saturated or partially unsaturated cyclic group having 3 to 6 ring carbon atoms, i.e. 3, 4, 5, or 6 carbon atoms. A carbocycle is preferably a “cycloalkyl”, which as used herein refers to a fully saturated hydrocarbon cyclic group. Preferably, a cycloalkyl group is a C₃-C₆ cycloalkyl group.

[0116] The term “C₆₋₁₀ aryl group” used herein means an aryl group constituted by 6, 7, 8, 9 or 10 carbon atoms and includes condensed ring groups such as monocyclic ring group, or bicyclic ring group and the like. Specifically, examples of “C₆₋₁₀ aryl group” include phenyl group, indenyl group, naphthyl group or azulenyl group and the like. It should be noted that condensed rings such as indan and tetrahydronaphthalene are also included in the aryl group.

[0117] The terms “alkylaryl” as used herein refers to an alkyl group as defined below substituted with an aryl as defined above. The alkyl component of an “alkylaryl” group may be substituted with any one or more of the substituents listed above for an aliphatic group and the aryl or heteroaryl component of an “alkylaryl” or “alkylheteroaryl” group may be substituted with any one or more of the substituents listed above for aryl, and carbocycle groups. Preferably, alkylaryl is benzyl.

[0118] The term “heteroaryl” as used herein refers to a monocyclic or bicyclic aromatic ring system having from 5 to 10 ring atoms, i.e. 5, 6, 7, 8, 9, or 10 ring atoms, at least one ring atom being a heteroatom selected from O, N or S.

[0119] An aliphatic, aryl, heteroaryl, or carbocycle group as referred to herein may be unsubstituted or may be substituted by one or more substituents independently selected from the group consisting of halo, C₁₋₆ alkyl, —NH₂, —NO₂, —SO₃H, —OH, alkoxy, —COOH, or —CN.

[0120] The term “halogen atom” or “halo” used herein means a fluorine atom, a chlorine atom, a bromine atom, an iodine atom and the like, preferably a fluorine atom or a chlorine atom, and more preferably a fluorine atom.

[0121] “C₂₋₆ Alkoxy” refers to the above C₁₋₆ alkyl group bonded to an oxygen that is also bonded to the cation ring. A “C₂₋₆ alkoxyalkyl group” refers to an alkyl containing an ether group, with the general formula X—O—Y wherein X and Y are each independently a C₁₋₅ alkyl and the total number of carbon atoms is between 2 and 6, e.g., 2, 3, 4, 5, or 6.

[0122] As used herein the term “alkenyl” refers to a linear or branched alkenyl group or moiety containing from 2 to 20 carbon atoms, such as 11, 12, 13, 14, 15, 16, 17, 18, or 19 carbon atoms. Preferably the alkenyl group or moiety contains 2-10 carbon atoms, i.e., 2, 3, 4, 5, 6, 7, 8, 9, or 10 carbon atoms such as a C₂₋₄ alkenyl or a C₂₋₆ alkenyl group or moiety, for example ethenyl, 1-propenyl, 2-propenyl, 1-butenyl, 2-butenyl, 3-butenyl, 1-pentenyl, 2-pentenyl, 3-pentenyl, 4-pentenyl, 1-hexenyl, 2-hexenyl, 3-hexenyl, 4-hexenyl, and 5-hexenyl.

[0123] As used herein the term “alkynyl” refers to a linear or branched alkynyl group or moiety containing from 2 to 20 carbon atoms, such as 11, 12, 13, 14, 15, 16, 17, 18, or 19 carbon atoms. Preferably the alkynyl group or moiety contains 2-10 carbon atoms i.e. 2, 3, 4, 5, 6, 7, 8, 9, or 10 carbon atoms such as a C₂₋₄ alkynyl or a C₂₋₆ alkynyl group or moiety, for example ethynyl, 1-propynyl, 2-propynyl, 1-butyne, 2-butyne, 3-butyne, 1-pentyne, 2-pentyne, 3-pentyne, 4-pentyne, 1-hexyne, 2-hexyne, 3-hexyne, 4-hexyne, and 5-hexyne.

[0124] The disclosure provides aspects and embodiments as set out in the following clauses:

[0125] 1. A method for making fibres, the method comprising:

[0126] providing a spinning dope comprising a dope solvent, lignin dissolved in the dope solvent and an additive polymer dissolved in the dope solvent, wherein the dope solvent comprises an ionic liquid and water; and extruding the spinning dope into a coagulant to obtain one or more fibres.

[0127] 2. The method of clause 1, wherein the water content in the dope solvent is at least 5 wt %.

[0128] 3. The method of clause 1 or 2, wherein the water content in the dope solvent is 5-40 wt %.

[0129] 4. The method of any preceding clause, wherein the water content in the dope solvent is 10-40 wt %.

[0130] 5. The method of any preceding clause, wherein the water content in the dope solvent is 20-40 wt %.

[0131] 6. The method of any preceding clause, wherein the spinning dope has a total loading of lignin and additive polymer of 6-60 wt % or 6-50 wt %, relative to the mass of mass of spinning dope, excluding the mass of lignin and additive polymer.

[0132] 7. The method of clause 6, wherein the total loading of lignin and additive polymer is 6-40 wt %.

[0133] 8. The method of clause 6, wherein the total loading of lignin and additive polymer is 6-30 wt %.

[0134] 9. The method of clause 6, wherein the total loading of lignin and additive polymer is 8-23 wt %.

[0135] 10. The method of clause 6, wherein the total loading of lignin and additive polymer is 11-20 wt %.

[0136] 11. The method of clause 6, wherein the total loading of lignin and additive polymer is 10-20 wt %.

[0137] 12. The method of clause 6, wherein the total loading of lignin and additive polymer is 15-20 wt %.

[0138] 13. The method of any preceding clause, wherein the weight ratio of lignin to additive polymer in the spinning dope is at least 2:1.

[0139] 14. The method of clause 13, wherein the weight ratio of lignin to additive polymer in the spinning dope or at least 2.5:1.

[0140] 15. The method of clause 14, wherein the weight ratio of lignin to additive polymer in the spinning dope or at least 3:1.

[0141] 16. The method of clause 13, wherein the weight ratio of lignin to additive polymer in the spinning dope is from 2:1 to 15:1.

[0142] 17. The method of clause 16, wherein the weight ratio of lignin to additive polymer in the spinning dope is from 2:1 to 10:1.

[0143] 18. The method of clause 17, wherein the weight ratio of lignin to additive polymer in the spinning dope is from 2.5:1 to 10:1.

[0144] 19. The method of clause 18, wherein the weight ratio of lignin to additive polymer in the spinning dope 3:1 to 9:1.

[0145] 20. The method of any preceding clause, wherein lignin is present in the spinning dope at a loading of at least 5 wt %, relative to the mass of the spinning dope, excluding the mass of lignin and additive polymer.

[0146] 21. The method of clause 20, wherein lignin is present in the spinning dope at a loading of at least 7 wt %.

[0147] 22. The method of clause 20, wherein lignin is present in the spinning dope at a loading of at least 10 wt %.

[0148] 23. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 5-50 wt %.

[0149] 24. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 5-40 wt %.

[0150] 25. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 5-30 wt %.

[0151] 26. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 5-20 wt %.

[0152] 27. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 5-15 wt %.

[0153] 28. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 7-30 wt %.

[0154] 29. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 7-20 wt %.

[0155] 30. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 7-15 wt %.

[0156] 31. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 10-50 wt %.

[0157] 32. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 10-40 wt %.

[0158] 33. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 10-30 wt %.

[0159] 34. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 10-20 wt %.

[0160] 35. The method of clause 20, wherein lignin is present in the spinning dope at a loading of 10-15 wt %.

[0161] 36. The method of any preceding clause, wherein the additive polymer is present in the spinning dope at a loading of 1-10 wt %, relative to the mass of spinning dope, excluding the mass of lignin and additive polymer.

[0162] 37. The method of clause 36, wherein the additive polymer loading is 1-5 wt %.

[0163] 38. The method of any preceding clause, wherein the spinning dope has a water content of at least 5 wt %, calculated based on the mass of water present relative to the mass of the spinning dope, excluding the mass of lignin and additive polymer.

[0164] 39. The method of any preceding clause, wherein the spinning dope has a water content of 5-40 wt %, calculated based on the mass of water present relative to the mass of the spinning dope, excluding the mass of lignin and additive polymer.

[0165] 40. The method of any preceding clause, wherein the spinning dope has a water content of 5-40 wt %, calculated based on the mass of water present relative to the mass of the spinning dope, excluding the mass of lignin and additive polymer.

[0166] 41. The method of any preceding clause, wherein the spinning dope has a water content of 10-40 wt %, calculated based on the mass of water present relative to the mass of the spinning dope, excluding the mass of lignin and additive polymer.

[0167] 42. The method of any preceding clause, wherein the spinning dope has a water content of 20-40 wt %, calculated based on the mass of water present relative to the mass of the spinning dope, excluding the mass of lignin and additive polymer.

[0168] 43. The method of any preceding clause, wherein the additive polymer is selected from poly(vinyl alcohol) (PVA), polyvinyl acetate, polyfurfuryl alcohol, polyacrylic acid, polyethylene oxide, polyethylene imine, poly(2-hydroxyethyl methacrylate), polyoxymethylene and mixtures thereof.

[0169] 44. The method of any preceding clause, wherein the additive polymer is PVA.

[0170] 45. The method of clause 44, wherein the PVA is partially hydrolysed, having a degree of hydrolysis (DH) of 72% or higher.

[0171] 46. The method of clause 44, wherein the PVA is partially hydrolysed, having a degree of hydrolysis (DH) of 72-95%.

[0172] 47. The method of clause 44, wherein the PVA is partially hydrolysed, having a degree of hydrolysis (DH) of 80-95%.

[0173] 48. The method of clause 44, wherein the PVA is partially hydrolysed, having a degree of hydrolysis (DH) of 85-90%.

[0174] 49. The method of clause 44, wherein the PVA is partially hydrolysed, having a degree of hydrolysis (DH) of 86.7-88.7%.

[0175] 50. The method of any preceding clause, wherein the weight average molecular weight (Mw) of the additive polymer is from 60 to 200 kDa, preferably 80 to 190 kDa.

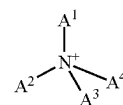
[0176] 51. The method of any preceding clause, wherein the weight average molecular weight (Mw) of the additive polymer is from 80 to 190 kDa.

[0177] 52. The method of any preceding clause, wherein the ionic liquid comprises a cation and an anion, wherein the anion is selected from C_{1-20} alkyl sulfate ($[AlkylSO_4]^-$), C_{1-20} alkylsulfonate ($[AlkylSO_3]^-$), hydrogen sulfate ($[HSO_4]^-$), hydrogen sulfite ($[HSO_3]^-$), dihydrogen phosphate ($[H_2PO_4]^-$), hydrogen phosphate ($[HPO_4]^{2-}$), chloride (Cl^-), bromide (Br^-), trifluoromethanesulfonate ($[OTf]^-$), formate ($[HCOO]^-$) and acetate ($[MeCO_2]^-$).

[0178] 53. The method of clause 52, wherein the anion is selected from $[HSO_4]^-$ and $[HCOO]^-$.

[0179] 54. The method of any preceding clause, wherein the ionic liquid comprises a cation and an anion, wherein the cation is a protic cation.

[0180] 55. The method of any preceding clause, wherein the ionic liquid comprises a cation and an anion, wherein the cation contains a nitrogen-containing heterocyclic moiety or wherein the cation is a cation of Formula I



[0181] wherein A^1 to A^4 are each independently selected from H, an aliphatic, C_{3-6} carbocycle, C_{6-10} aryl, alkylaryl, and heteroaryl.

[0182] 56. The method of any preceding clause, wherein the ionic liquid is an [alkylammonium][HSO_4] or [alkylammonium][$HCOO$] ionic liquid.

[0183] 57. The method of any preceding clause, wherein the ionic liquid is triethylammonium hydrogen sulfate $[TEA][HSO_4]$, N,N-dimethylbutylammonium hydrogen sulfate $[DMBA][HSO_4]$, diethylammonium hydrogen sulfate $[DEA][HSO_4]$, N,N-dimethylethylammonium hydrogen sulfate $[DMEA][HSO_4]$, diethanolammonium chloride $[DEtOHA]Cl$, 1-methylimidazolium chloride $[HMim]Cl$, 1-ethyl-3-methylimidazolium chloride $[EMim]Cl$, and 1-ethyl-3-methylimidazolium trifluoromethanesulfonate $[EMim][OTf]$, 1-butylimidazolium hydrogen sulfate $[HBim][HSO_4]$, methylbutylammonium hydrogen sulfate

([MBA][HSO₄]), 1-methylimidazolium formate ([HMim][HCOO]), N,N-dimethylbutylammonium formate ([DMBA][HCOO]), N,N-dimethylethylammonium hydrogen sulfate ([DMEA][HSO₄]), 1-butyl-3-methylimidazolium hydrogen sulfate [BMim][HSO₄], or N,N-dimethylbutylammonium acetate ([DMBA][OAc]), or a mixture thereof.

[0184] 58. The method of any preceding clause, wherein the ionic liquid is N,N-dimethylethylammonium hydrogen sulfate ([DMEA][HSO₄]), N,N-dimethylbutylammonium hydrogen sulfate ([DMBA][HSO₄]), 1-butylimidazolium hydrogen sulfate ([HBim][HSO₄]), triethylammonium hydrogen sulfate ([TEA][HSO₄]), methylbutylammonium hydrogen sulfate ([MBA][HSO₄]), 1-methylimidazolium formate ([HMim][HCOO]), N,N-dimethylbutylammonium formate ([DMBA][HCOO]), 1-methylimidazolium hydrogen chloride [HMim]Cl, N,N-dimethylbutylammonium chloride [DMBA]Cl, 1-butyl-3-methylimidazolium hydrogen sulfate [BMim][HSO₄], or N,N-dimethylbutylammonium acetate ([DMBA][OAc]), or a mixture thereof.

[0185] 59. The method of any preceding clause, wherein the ionic liquid is N,N-dimethylbutylammonium hydrogen sulfate ([DMBA][HSO₄]), 1-butylimidazolium hydrogen sulfate ([HBim][HSO₄]), triethylammonium hydrogen sulfate ([TEA][HSO₄]), methylbutylammonium hydrogen sulfate ([MBA][HSO₄]), 1-methylimidazolium formate ([HMim][HCOO]), N,N-dimethylbutylammonium formate ([DMBA][HCOO]), 1-methylimidazolium hydrogen chloride [HMim]Cl, N,N-dimethylbutylammonium chloride [DMBA]Cl, 1-butyl-3-methylimidazolium hydrogen sulfate [BMim][HSO₄], or N,N-dimethylbutylammonium acetate ([DMBA][OAc]), or a mixture thereof.

[0186] 60. The method of any preceding clause, wherein the ionic liquid is N,N-dimethylbutylammonium hydrogen sulfate ([DMBA][HSO₄]), 1-butylimidazolium hydrogen sulfate ([HBim][HSO₄]), triethylammonium hydrogen sulfate ([TEA][HSO₄]), methylbutylammonium hydrogen sulfate ([MBA][HSO₄]), 1-methylimidazolium formate ([HMim][HCOO]), or N,N-dimethylbutylammonium formate ([DMBA][HCOO]), or a mixture thereof.

[0187] 61. The method of any preceding clause, wherein the ionic liquid is [DMBA]Cl, [DMBA][HSO₄], [BMim][HSO₄], [MBA][HSO₄] or [HBim][HSO₄], or a mixture thereof.

[0188] 62. The method of any preceding clause, wherein the ionic liquid is [DMBA][HSO₄].

[0189] 63. The method of any preceding clause, wherein the dope solvent further comprises ethanol.

[0190] 64. The method of clause 63, wherein the ethanol is present at 1-20 wt %, relative to the total weight of dope solvent.

[0191] 65. The method of clause 64, wherein the ethanol is present at 5-15 wt %.

[0192] 66. The method of any of clauses 63 to 65, wherein the ionic liquid:ethanol mass ratio is from 3:1 to 15:1.

[0193] 67. The method of clause 66, wherein the ionic liquid:ethanol mass ratio is from 5:1 to 12:1.

[0194] 68. The method of any preceding clause, wherein the coagulant comprises water.

[0195] 69. The method of any preceding clause, wherein the coagulant comprises water and ionic liquid, wherein the ionic liquid is present at no more than 60 wt %, relative to the total mass of coagulant.

[0196] 70. The method of any preceding clause, wherein the coagulant comprises water and ionic liquid, wherein the ionic liquid is present at no more than 30 wt %, relative to the total mass of coagulant.

[0197] 71. The method of any preceding clause, wherein the coagulant comprises water and ionic liquid, wherein the ionic liquid is present at no more than 15 wt %, relative to the total mass of coagulant.

[0198] 72. The method of any preceding clause, wherein the coagulant comprises water and ionic liquid, wherein the ionic liquid is present at 1-60 wt %, relative to the total mass of coagulant.

[0199] 73. The method of any preceding clause, wherein the coagulant comprises water and ionic liquid, wherein the ionic liquid is present 1-30 wt %, relative to the total mass of coagulant.

[0200] 74. The method of any preceding clause, wherein the coagulant comprises water and ionic liquid, wherein the ionic liquid is present at 1-15 wt %, relative to the total mass of coagulant.

[0201] 75. The method of any preceding clause, wherein the coagulant comprises water and ionic liquid, wherein the ionic liquid is present at 5-15 or 5-10 wt %, relative to the total mass of coagulant.

[0202] 76. The method of any preceding clause, wherein the coagulant comprises water and sodium sulfate.

[0203] 77. The method clause 76, wherein the coagulation bath comprises aqueous sodium sulfate at a concentration of 0.5-1.5 M.

[0204] 78. The method of any preceding clause, wherein the spinning dope is prepared by a process comprising: a) providing the ionic liquid, optionally in a mixture with water, adding an aqueous solution of the additive polymer and, optionally, adding water to reach the desired water content to provide a solution of additive polymer in dope solvent; and b) adding lignin to the solution formed in step a) to dissolve the lignin and provide the spinning dope.

[0205] 79. The method of clause 78, wherein dissolution of the lignin is performed at a temperature of 10° C. to 200° C., preferably 10° C. to 100° C.

[0206] 80. The method of clause 78, wherein dissolution of the lignin is performed at a temperature of, more preferably 20° C. to 100° C., 20° C. to 60° C. or 20° C. to 30° C.

[0207] 81. The method of any preceding clause, further comprising aging the spinning dope for at least 2 minutes prior to the extruding.

[0208] 82. The method of any preceding clause, further comprising aging the spinning dope for at least 5 minutes prior to the extruding.

[0209] 83. The method of any preceding clause, further comprising aging the spinning dope for at least 30 minutes prior to the extruding.

[0210] 84. The method of any preceding clause, further comprising aging the spinning dope for 6-48 hours prior to the extruding.

[0211] 85. The method of any preceding clause, further comprising aging the spinning dope for 6-24 hours prior to the extruding.

[0212] 86. The method of any of clauses 81 to 85, wherein the spinning dope is heated during aging at a temperature of 20° C. to 90° C.

[0213] 87. The method of any of clauses 81 to 85, wherein the spinning dope is heated during aging at a temperature of 30° C. to 90° C.

[0214] 88. The method of any of clauses 81 to 85, wherein the spinning dope is heated during aging at a temperature of 30° C. to 60° C.

[0215] 89. The method of any of clauses 81 to 85, wherein the spinning dope is heated during aging at a temperature of at least 30° C.

[0216] 90. The method of any of clauses 81 to 85, wherein the spinning dope is heated during aging at a temperature of at least 60° C.

[0217] 91. The method of any of clauses 81 to 90, wherein the spinning dope is heated during aging at a temperature of up to 150° C.

[0218] 92. The method of any of clauses 86 to 91, wherein the spinning dope is heated for at least 2 minutes.

[0219] 93. The method of any of clauses 86 to 91, wherein the spinning dope is heated for at least 5 minutes.

[0220] 94. The method of any of clauses 86 to 91, wherein the spinning dope is heated for at least 30 minutes.

[0221] 95. The method of any of clauses 86 to 91, wherein the spinning dope is heated for 30 minutes to 6 hours.

[0222] 96. The method of any of clauses 86 to 91, wherein the spinning dope is heated for 1 to 10 hours.

[0223] 97. The method of any of clauses 81 to 96, wherein the spinning dope is mixed prior to the extruding for at least 2 minutes.

[0224] 98. The method of any of clauses 81 to 96, wherein the spinning dope is mixed prior to the extruding for at least 5 minutes.

[0225] 99. The method of any of clauses 81 to 96, wherein the spinning dope is mixed prior to the extruding for at least 30 minutes.

[0226] 100. The method of any of clauses 81 to 96, wherein the spinning dope is mixed prior to the extruding for 30 minutes to 10 hours.

[0227] 101. The method of any of clauses 81 to 96, wherein the spinning dope is mixed prior to the extruding for 1 to 6 hours.

[0228] 102. The method of any preceding clause, wherein the providing the spinning dope comprises:

[0229] a) contacting a lignocellulosic biomass comprising lignin and cellulose with a composition comprising the ionic liquid and water, to dissolve the lignin and produce a cellulose pulp;

[0230] b) separating the cellulose pulp to obtain a liquor comprising the ionic liquid, water and lignin; and

[0231] c) combining the liquor with the additive polymer to obtain the spinning dope.

[0232] 103. The method of clause 102 wherein step c) of combining the liquor with the additive polymer may comprise combining the liquor with an aqueous solution of the additive polymer.

[0233] 104. The method of clause 102, wherein the composition comprising the ionic liquid and water has a 5-40 wt % water content.

[0234] 105. The method of clause 102, wherein the composition comprising the ionic liquid and water has a 5-30 wt % water content.

[0235] 106. The method of clause 102, wherein the composition comprising the ionic liquid and water has a 10-30 wt % water content.

[0236] 107. The method of any of clauses 102-106, wherein the lignocellulosic biomass contacted with the composition is heated to 100-180° C.

[0237] 108. The method of any of clauses 102-106, wherein the lignocellulosic biomass contacted with the composition is heated to 120-170° C.

[0238] 109. The method of any of clauses 102-106, wherein the lignocellulosic biomass contacted with the composition is heated to 120-150° C.

[0239] 110. The method of any of clauses 102-109, wherein the lignocellulosic biomass is contacted with the composition for 1 minute to 22 hours.

[0240] 111. The method of any of clauses 102-109, wherein the lignocellulosic biomass is contacted with the composition for 10 minutes to 22 hours.

[0241] 112. The method of any of clauses 102-109, wherein the lignocellulosic biomass is contacted with the composition for 10 minutes to 10 hours.

[0242] 113. The method of any of clauses 102-109, wherein the lignocellulosic biomass is contacted with the composition for 15 minutes to 8 hours.

[0243] 114. The method of any of clauses 102-109, wherein the lignocellulosic biomass is contacted with the composition for 30 minutes to 8 hours.

[0244] 115. The method of any preceding clause, further comprising washing the one or more fibres after extrusion, optionally wherein the washing is carried out with water.

[0245] 116. The method of any preceding clause, further comprising drying the one or more fibres.

[0246] 117. The method of any preceding clause, further comprising drying the one or more fibres under mechanical tension.

[0247] 118. The method of any preceding clause, further comprising heating the one or more fibres in air.

[0248] 119. The method of clause 118, comprising heating the one or more fibres in air at 150-300° C.

[0249] 120. The method of clause 118 or 119, further comprising weaving the fibres to form a fabric.

[0250] 121. The method of any preceding clause, further comprising carbonising the one or more fibres to obtain carbon fibres.

[0251] 122. The method of clause 121, wherein carbonising comprises heating the one or more fibres to 800-3000° C., under an inert atmosphere.

[0252] 123. The method of clause 122, wherein carbonising comprises heating the one or more fibres to 1200-1800° C., under an inert atmosphere.

[0253] 124.

[0254] The method of any of clauses 121-123, wherein carbonising is carried out on the fibres under tension.

[0255] 125. The method of any preceding clause, wherein the spinning dope comprises a cellulose loading of no more than 10 wt %, relative to the mass of spinning dope, excluding the mass of cellulose, lignin and additive polymer.

[0256] 126. The method of any preceding clause, wherein the spinning dope comprises a cellulose loading of no more than 5 wt %.

[0257] 127. The method of any preceding clause, wherein the spinning dope comprises a cellulose loading of no more than 4 wt %.

[0258] 128. The method of any preceding clause, wherein the spinning dope comprises a cellulose loading of no more than 1 wt %.

[0259] 129. A fibre obtainable by the method of any of the preceding clauses.

[0260] 130. A fibre obtained by the method of any of the preceding clauses.

[0261] 131. A fabric comprising one or more fibres according to clause 129 or clause 130.

[0262] The present invention will now be described by way of reference to the following examples and accompanying drawings which are present for the purposes of illustration only and are not to be construed as being limiting on the invention.

EXAMPLES

Materials and Methods

[0263] Willow (variety Endurance) wood was obtained from Agri-food & Biosciences Institute, UK, and *Eucalyptus* (*Eucalyptus grandis*, also Red *grandis*) wood was purchased from W. L. West & Sons Ltd. The willow and *Eucalyptus* biomass were milled to a particle size below 1 cm with a cutting mill (Retch SM 2000) and stored in a plastic bag to protect from sunlight. The moisture content of the biomass was determined in triplicate by measuring the loss of mass during drying at 105° C. for at least 24 hours.

[0264] Softwood kraft lignins were obtained from Research Institute of Sweden (mentioned as Softwood Kraft lignin 1) and Sigma Aldrich (Lignin Alkali, mentioned as Softwood Kraft lignin 2). N,N-dimethylbutylamine (purity >99%) was purchased from Sigma Aldrich and 66.3% sulfuric acid solution was purchased from VWR. Poly(vinyl alcohol) (PVA) was purchased from Kuraray Poval (No. 13-88, 88% hydrolysed, ca. 100 kDa). All reagents were used as received.

Optical Microscopy (OM)

[0265] Optical microscope images of the dope were taken on DM2500 microscope with a Basler Ace acA1920 camera. The sample was prepared by drawing solution into a syringe through the needle, followed by depositing a small drop on a glass slide from the needle. The sample was covered with a cover glass and imaged. Crossed polarisers were used to identify birefringent particles.

Viscosity

[0266] The viscosity of the Lignin/PVA solutions was measured using an AR 2000ex rheometer with a cone-and-plate feature (2° cone angle, 20 mm plate diameter and 53 µm gap) at shear rates from 1 to 1000 s⁻¹ at room temperature. Steady state flow measurements were carried out to determine the effect of increasing the shear rate on the dope solutions. Each measurement was held at a point time of 30 s.

Scanning Electron Microscope (SEM)

[0267] SEM images of the lignin-PVA fibres were recorded on a JEOL JSM-6010LA. The fibres were broken into pieces with tweezers and mounted on an aluminium support using carbon adhesive tape. Samples were sputter coated with chromium before imaging. The accelerating voltage used was 20 kV.

Tensile Testing

[0268] The tensile strength and stiffness of the Lignin-PVA precursor and carbon fibres samples were determined using the standard test method ISO BSI11566. The single filaments were mounted on a card template (15 mm gauge)

using an epoxy adhesive (Araldite Rapid, Huntsman Corporation, US). Samples were tested using a tensile tester (Linkam Scientific Ltd. GB) fitted with a 20 M loading cell at 16.7 µm s⁻¹ until failure. The tensile modulus was determined from linear regions between 0.2% and 0.5% strain. The cross-sectional areas of the samples were measured using an optical microscope.

Thermogravimetric Analysis (TGA)

[0269] Thermogravimetric analysis (TGA) was performed on a thermogravimetric analyser (Mettler Toledo TGA/DSC 1LF/UMX) to predict the carbon yield for the precursor fibres. For determining carbon yield of precursor fibres, the samples were heated in platinum pans at 10° C. min⁻¹ flow of nitrogen from 25° C. to 100° C., held isothermally at 100° C. for 30 min to drive off moisture and the temperature was ramped to 900° C. at 10° C. min⁻¹. Weight loss during thermal stabilisation was also determined using a thermogravimetric analyser (Mettler Toledo TGA/DSC 1LF/UMX). For carbonisation, the samples were heated in platinum pans at 10° C. min⁻¹ flow of nitrogen from 25° C. to 100° C., held isothermally at 100° C. for 30 min to drive off moisture and the temperature was ramped to 900° C. at 10° C. min⁻¹. For the thermal stabilisation, fibre samples were heated in air from 25 to 100° C. at 1° C./min, and from 100 to 250° C. at 0.2° C./min, followed by holding at 250° C. for 1 h. Air/N₂ flow rate was 50 mL/min.

Raman Spectroscopy

[0270] The graphitic order in lignin-derived CFs was investigated through Raman spectroscopy using a Renishaw in Via micro-Raman spectrometer with 532 nm (2.33 eV) DPSS diode laser. Raman spectra in the range of 80-3200 cm⁻¹ were collected using software WiRE 4.1 and averaged for at least five different locations along the CF surface. The intensity ratio of G-(1582 cm⁻¹) to D-mode (1350 cm⁻¹) (IG/ID) was measured after fitting D- and G-band with a Lorentzian fitting using WiRE 4.1 software.

Example 1. Ionic Liquid Synthesis

[0271] The ionic liquid [DMBA][HSO₄] was synthesised from N,N-dimethylbutylamine and a 66.3% sulfuric acid solution in a custom-built flow reactor. All reagents were used as received. The precursors were chilled and pumped into a stirred flow reactor at flow rates of 5 ml/min for the acid and 7.8 ml/min for the base. The acid/base ratio of the produced IL was checked in triplicate using an automatic titrator (Mettler Toledo G205), where the HSO₄-anion was titrated with aqueous NaOH. For the correction of the acid/base ratio, a calculated amount of dimethylbutylamine or 66.3% sulfuric acid was gradually added to the IL cooled in an ice bath. The water content of the IL was adjusted as required: it was first reduced by a rotary evaporator and then measured in triplicate using a volumetric Karl Fischer titrator (Mettler Toledo V20). The final water content of the IL was confirmed with a Karl Fischer volumetric titrator.

Example 2. IonoSolv Lignin Extraction

[0272] Lignin extraction was carried out following the IonoSolv pretreatment procedure (Gschwend, F. J. v. et al. *Journal of Visualized Experiments* 2016, 2016 (114), 4-9). 30 g (dry-basis) biomass was added into a 200 mL pressure tube (Ace Glass pressure tube, front sealing), followed by

adding 150 g of [DMBA][HSO₄]_{80%}/water_{20%}. The biomass and IL were well-mixed with a vortex shaker (VWR) until all biomass particles were in contact with the IL. The pressure tube was placed in a preheated oven for 1 h at 170° C. After the pretreatment, the mixture was transferred from the pressure tube into a 1000 mL glass bottle, followed by mixing with 600 mL absolute ethanol (EtOH), shaken well and left to rest for 1 h at room temperature. After 1 h, the mixture was separated into a cellulose rich solid and a liquid containing ionic liquid, water, ethanol and the dissolved biomass components including the lignin (liquor) using vacuum filtration. The cellulose was air-dried. The washing steps were repeated three times. Each time, the liquor was collected. The majority of the water contained in the IL solvent and EtOH was evaporated from the combined liquor fraction with a rotary evaporator. To precipitate the lignin, 400 mL of deionised water (acting as an anti-solvent) was added to the reconcentrated liquor and the mixture was transferred into a 500 mL centrifuge tube (Corning®), shaken well using a vortex shaker and left to rest for 1 h before centrifugation (Megastar 3.0, VWR, UK; 3000 rpm, 50 min). The supernatant was decanted. After washing the precipitate with deionised water for 3 times, the lignin was freeze-dried for 2 days. The dried lignin and the air-dried cellulose pulp were weighed on aluminium foil to determine lignin and pump yields (moisture content of pulp was measured separately and subtracted from the air-dried yield). The fractionation was carried out in triplicate.

Example 3. Preparation of Spinning Dope

[0273] The poly(vinyl alcohol) stock solution, PVA_(aq), and spinning dopes were prepared on the day of the fibre spinning experiment. The weight fraction of PVA in PVA_(aq) was dependent on the desired solid loading, water content and lignin/PVA weight ratio in the dope, and the water content in the ionic liquid. The required PVA loading in deionised (D.I.) water for the preparation of PVA_(aq) was calculated using equation (1):

$$W_{p/w} = \frac{W_{sl/sv}(1 - W_{w,IL})}{(1 + W_{1/p})(W_{w,sv} - W_{w,IL})} \quad (1)$$

[0274] where $W_{p/w}$ =Weight ratio of PVA to water in PVA solution (i.e. PVA loading); $W_{sl/sv}$ =desired weight ratio of solid to solvent (solid loading); W_{vp} =desired weight ratio of lignin to PVA; $W_{w,IL}$ =water fraction of water in the IL; $W_{w,sv}$ =desired weight fraction of water in the solvent. Note that the solvent here is the mixture of IL and water with desired water content.

[0275] The amount of PVA_(aq) solution and lignin to be added into the IL was calculated according to the equation (2) and (3) below:

$$W_{PVA_{aq}} = \frac{W_{IL}(1 + W_{p/w})(W_{w,sv} - W_{w,IL})}{(1 - W_{w,sv})} \quad (2)$$

[0276] where $W_{PVA_{aq}}$ =weight of PVA(aq) to be added and W_{IL} =weight of IL added;

$$W_{lig} = \frac{W_{PVA_{aq}} W_{1/p} W_{p/w}}{1 + W_{p/w}} \quad (3)$$

[0277] where W_{lig} =weight of lignin to be added

[0278] To prepare the PVA solution, PVA flakes were slowly added into a pre-weighed 50 ml round bottom flask with stir bar followed by deionised water at room temperature while stirring. The solution was mixed for 15 min at room temperature, followed by increasing the temperature to 85° C. and stirring for 1 h until the polymer flakes had dissolved. The round bottom flask was weighed again to monitor the loss of water. To prepare a spinning dope with 16% solid loading (lignin/PVA=3:1), for example (according to equation (2) and (3)), 1.5 g of IL (water content of 4 wt %) was added into a 10 mL round bottom flask with stir bar. 1 g of freshly prepared PVA_(aq) solution (10.67 wt %) was added into the IL and mixed while stirring at 700 rpm on a hotplate at 60° C. for 5 min. 0.288 g of lignin was added slowly into the PVA/IL/water mixture and the solution mixed for 1 h at 60° C. Dopes with three lignin/PVA ratios were prepared to have different lignin content in the fibres (see Table 1).

TABLE 1

Lignin and PVA content in the dopes (mass of each polymer relative to the mass of dope solvent) and corresponding lignin content in the fibre		
Lignin content in the fibre	Lignin content in dope	PVA content in dope
75%	12.0%	4.0%
82.5%	13.2%	2.8%
90%	14.4%	1.6%

Example 4. Fibre Spinning

[0279] The dope solutions were aspirated into a using a 1 mL plastic syringe without needle and added into 1 mL glass syringes (Hamilton® syringe, 1000 series GASTIGHT®, PTFE luer lock) via the plunger end with the plunger removed. The plunger was inserted, and the syringe held vertically with the plunger at the bottom to remove all air bubbles through the opening in the top while carefully pressing the plunger. A Z-shaped 27-gauge (internal diameter 0.21 mm) cannula (Central Surgical UK) was attached to the syringe. The dope solution (ca. 0.5 mL) was extruded with a syringe pump (KDS LEGATO™ 100, KD Scientific, US) into the coagulation bath (1M Na₂SO₄ solution or deionised water) which was placed on a rotational table at 9 RPM. The extrusion rate was 0.6 mL/h or 1.5 mL/hL. The bath was rotated at 7.5 cm/s at the point of injection to provide extensional flow during fibre coagulation. The theoretical ratio of rotating speed of the coagulant fluid (at the point of injection) to the injection speed was 15.6. However, due to the limited viscous force of coagulant, the drawing is limited and did not reach to the theoretical draw ratio was not reached, as fibres were pulled into the centre of the bath, where the rotation speed was much lower. The fibres were soaked for 30 minutes in the coagulation bath containing 1M Na₂SO₄ or water. Fibre coagulated in 1 M Na₂SO₄ were transferred into a separate water bath for 30 min for washing (this step is not needed when coagulating fibres in water).

The fibres were hung on a rod under tension from a known weight that was attached to the bottom end of the fibres (around 4 mN) and air dried at room temperature (25° C.) overnight.

Example 5. Stabilisation and Carbonisation

[0280] For thermal stabilisation, the air-dried lignin-PVA precursor fibres were hung vertically under tension (ca. 5.5 mN weight) during thermal stabilisation. The precursor fibres were stabilised in a convection oven (Memmert UNP-200) with air as the gas atmosphere by heating from 25 to 100° C. at 1° C./min, and from 100 to 250° C. at 0.2° C./min, followed by holding at 250° C. for 1 h. Carbonisation was carried out in a quartz tube furnace (OTF-1200X-III-S-NT) under N₂ flow (50 mL/min), by heating from 25 to 1000° C. at 1° C./min, followed by a hold at 1000° C. for 2 h.

Example 6. Effect of Dope Preparation Temperature and Fibre Extrusion Rate

[0281] In general, dopes for lignin fibre spinning are prepared at a temperature range of room temperature to 60° C. In this example, dopes were prepared at three different temperatures (30° C., 45° C. and 60° C.). Dopes are homogeneous regardless of the preparation temperatures. Dope preparation temperature affects dope viscosity, and this effect was regardless of lignin types. The viscosity of dopes containing *Eucalyptus* ionoSolv lignin (extracted and isolated in the lab) or commercial softwood kraft lignin were measured. As can be seen in FIG. 1, when increasing the dope preparation temperature, dope viscosity increased. This was in contrast with the expectation that heating a solution would decrease the solution viscosity. This could be an indication of condensation reactions between lignin and PVA catalysed by the acidic environment (i.e. [DMBA] [HSO₄]_{60%}/water_{40%}) during dope preparation. The same increase in viscosity is not observed for control solutions such as lignin and PVA solutions.

[0282] During fibre spinning, the estimated shear rate experienced by the dope flowing through a 27 G needle at 0.6 ml/h is around 183 s⁻¹ according to equation 4 (assuming Newtonian behaviour).

$$\gamma_w = \frac{4Q}{\pi r^3} \quad (4)$$

[0283] where γ_w is the shear rate of the wall of the needle, Q is volumetric flow rate through the needle and r is the radius of the needle.

[0284] According to the rheology measurement (FIG. 1), at a shear rate of 183 s⁻¹, dopes containing *Eucalyptus* ionoSolv lignin (a hardwood lignin) experience observable shear-thinning whereas dope containing softwood kraft lignin experience Newtonian flow or only slight shear-thinning. This indicates that, at the applied extrusion rate, the more viscous ionoSolv lignin-containing dopes provide more polymer alignment, while for softwood kraft lignin-containing dopes less alignment was observed, as there was no measurable decrease in viscosity compared to lower shear rates. Increasing the extrusion rate to over 1.2 ml/h (eqv. to 366 s⁻¹) could promote more polymer alignment for softwood kraft lignin dopes during fibre spinning, which may be beneficial.

[0285] Stable fibre spinning was achieved at all tested temperatures (30, 45 & 60° C.) at different extrusion rates (0.6 ml/h and 1.5 ml/h) in Na₂SO₄ (aq), and little difference in fibre morphology was observed at when varying dope preparation temperatures and extrusion rate. All fibres extruded from dopes prepared at different temperatures exhibit smooth surface and circular cross-sections with uniform diameter along the fibre. An increase in fibre diameter was observed when increasing the dope preparation temperature (FIG. 2), which is due to the increased viscosity for dopes prepared at a higher temperature (FIG. 1). Increasing the extrusion rate also increased the fibre diameter (FIG. 2), as the effective draw ratio decreased.

[0286] A common strategy to increase fibre mechanical properties is by reducing fibre diameters, as this reduces the number of defects in each fibre, which introduce focal points for premature fracture under tension. In this study, although the fibre diameter decreased when preparing dopes at lower temperatures and extruding fibres at a lower extrusion rate, there was little change in the fibre tensile strengths (FIG. 3). This indicates that the fibre failure was not only defect-induced but also related to the microstructure. At the higher extrusion rate 1.5 ml/h, the calculated shearing rate of the dope flowing inside the needle is around 458 s⁻¹, suggesting that a more significant shear-thinning behaviour was experienced than during extruding at 0.6 ml/h (FIG. 1). At the higher extrusion rate, polymers align more during flowing inside the needle. The increased alignment may explain the offset of the strength loss caused by the increased fibres diameters.

[0287] When the dope was prepared at 60° C., the modulus of the precursor fibre increased regardless of extrusion rates (FIG. 3). The modulus of fibres extruded at 0.6 ml/h reached 4.5 GPa when the dope was prepared at 60° C. The increased viscosity of the dope due to the higher dope preparation temperature provided increased shear-thinning, as can be seen in FIG. 1. This promotes polymer alignment which improves the fibre modulus. Additionally, a high-viscosity dope may form denser fibres which can also result in an increase in fibre modulus. It was also observed in this study that extruding fibres at a lower rate improved the modulus, especially when the dope was prepared at a high temperature. Despite less shearing in the needle at a lower extrusion rate, the overall effect of low extrusion rate together with a higher dope preparation temperature was positive. For following studies, the dope preparation temperature was set to 60° C. and extrusion rate was set to 0.01 ml/min. Preparing dopes at 60° C. for 1 h was used for different lignin types (both ionoSolv and Kraft lignin).

Example 7. Effect of Lignin Content and Dope Aging Time

[0288] Maximising the lignin content in the precursor fibre should improve the carbon yield and therefore reduce the cost, as the carbon yield for lignin (~40%) is higher than the carbon yield of PVA (<10%). Dopes were prepared with lignin/PVA ratios from 3:1 to 9:1 (corresponding to 75-90% lignin content in the fibre, Table 1). Dopes were homogeneous at lignin/PVA ratios from 3:1 to 9:1, as confirmed by optical microscopy. The viscosity of the dopes (FIG. 4a) decreased with decreasing PVA content, as expected, as lignin molar weight is lower than PVA n (ionoSolv *Eucalyptus* lignin Mn ~1000 Da, Mw ~3000, willow lignin Mn~1000 Da, Mw~6000 Da) compared to the molar weight

of the PVA used (Mw ca. 100 kDa). For dopes containing 82.5% and 75% lignin (of total polymer content w/w), shear thinning was also observed when the shear rate was above 20 s^{-1} . At 90% lignin, the response was essentially Newtonian in the range studied. At 0.6 mL/h extrusion rate (eqv. to the region of 183 s^{-1} according to equation 4), the dope containing 75% and 82.5% lignin experienced shear-thinning flow with viscosities around 1.5 Pa·s, whereas the 90% lignin dope appeared to exhibit Newtonian behaviour with a viscosity of less than 0.5 Pa·s. Viscosities in a range of 0.5-6 Pa·s at a shear rate of ca. 183 s^{-1} (extrusion at 0.6 mL/h using a 27 G needle) was beneficial for stable and continuous spinnability. A viscosity of at least 0.6 Pa·s was beneficial for good handleability of fibres (avoiding breakage during picking up from the coagulation bath).

[0289] The *Eucalyptus* lignin-PVA spinning dopes were successfully spun into 1 M Na_2SO_4 (aq) coagulation bath, to generate well-formed gel proto-fibres. Good gel fibres were formed even at exceptionally high lignin content (90%). Continuous fibres containing 82.5% and 75% lignin were successfully removed from the bath and air-dried under tension.

[0290] Without being bound by theory, increase of dope viscosity with increasing dope preparation temperature, is proposed to occur due to a lignin/PVA reaction involving chemical cross-linking and hydrogen bonding between lignin and PVA in the acidic environment. It appears the reaction is irreversible. In this case, aging the dope should have the same effect on viscosity. Indeed, the dope viscosity increased over time even at room temperature (FIG. 4b). Within 6 h after the dope was prepared, the viscosity of dope had increased, with more intense shear thinning behaviour. At low shear rates, at which the dope exhibited Newtonian flow, the viscosity increased from 1.7 Pa·s to above 2 Pa·s after 6 h (FIG. 4), and further increased to 6 Pa·s after 24 h. The spinning of aged dopes was carried out to improve the strength of lignin fibres with very high lignin content (90%). In general, the 6 h- and 24 h-aged dopes did not affect the stability of the lignin/PVA fibre spinning. However, aging the spinning dope for 6 h or 24 h improved the ease of removing fibres containing 90% lignin content from the coagulation and water wash bath and allowed drying under tension. Observation of the ease of fibre spinning and drying is summarised in Table 2.

TABLE 2

Observations of lignin fibre spinning containing different amount of lignin			
Lignin in fibre (wt %)	Spinning	Coagulation and washing	Drying
90	Stable and continuous	Fibres collected from aged dope	6 h and 24 h-aged dope; dry under 0.2 mN (~20 mg)
82.5	Stable and continuous	Fibres collected from fresh/aged dope without breaking	Fresh and aged dope; dry under 4 mN (~400 mg)
75	Stable and continuous	Fibres collected from fresh/aged dope without breaking	Fresh and aged dope; dry under 4 mN (~400 mg)

[0291] Dried fibres had dense structures, with smooth surfaces, and circular cross-sections with uniform diameter as observed by SEM. In a previous study on spinning of

lignin/PVA fibres, bean-shaped fibres were obtained when the lignin content was 70%. All fibres produced in this manner had circular cross-sections, regardless of lignin content and lignin type, indicating the coagulation rate to be broadly appropriate. The fibres also appeared to have a homogenous microstructure without phase separation, on the fibre surface and inside, as evidence by inspecting the cross-sections. This is a sign of very good miscibility between lignin and PVA in the $[\text{DMBA}][\text{HSO}_4]_{60\%}/\text{water}_{40\%}$ dope solvent. Pores were observed in some fibre cross-sections, which is attributed to small air bubbles that had not been removed prior to dope extrusion. Degassing the dope to remove more air bubbles should minimize the number of pores in the fibres and improve their mechanical properties.

[0292] Dried fibres with a lignin content of 75% and 82.5% extruded in this way had similar diameters of around 100-120 μm , independent of aging time. The 90% lignin fibres had smaller fibre diameters of around 50 μm , which increased when the dope was aged. Given that the solid content of the dopes was constant, the change in fibre diameter must relate to the draw applied at the point of injection. The speed of the coagulation bath was 15.6 times higher than the linear injection rate of the dope, exerting an axial acceleration on the coagulating fibre. The more viscous (75 & 82.5% lignin, and aged 90% lignin) dopes resisted this acceleration more, limiting the drawing process, whilst more fluid dopes showed the greatest extension. The dope with a lignin content in solid=of 90% (relative to total polymer loading) had a lower viscosity, resulting in fibres with the thinnest diameter, whereas the dopes with 75% and 82.5% lignin content had similar viscosities, resulting in fibres with similar diameters.

[0293] The tensile modulus was consistent across the lignin fibres produced using this method, at around 4-5 GPa (FIG. 6c), indicating a similar density. The tensile strength (FIG. 6b) was the highest for the 75% lignin fibre at around 30-40 MPa. The 82% and 90% lignin fibres had lower strengths of around 25 MPa. Surprisingly, a similar or slightly higher strength was measured for 90% lignin fibres; an observation that may be attributed to their smaller diameter and hence reduced defect size. However, there is some scatter due to the tendency of the 90% lignin samples to break into shorter lengths during handling, and the variable tensioning weights used. The trends for the elongation at break (FIG. 6d) follow those for tensile strength.

Example 8. Effect of Lignin Type

[0294] To explore the generality of the approach, dopes were prepared with four lignins (two ionosolv hardwood lignins and two commercial softwood kraft lignins). Dopes containing *Eucalyptus* lignin were homogenous with no observable particles. The dope containing ionosolv willow lignin, contained uniformly distributed undissolved particles. Most particles are thought to be undissolved lignin, since they also appear in a control solution containing only lignin and dope solvent (12% lignin in $[\text{DMBA}][\text{HSO}_4]_{60\%}/\text{water}_{40\%}$). A smaller number of birefringent particles were observed under polarised light, likely residual cellulose crystals. Potentially these crystals could be removed by filtration in future. Two dopes containing Kraft softwood lignin showed complete lignin dissolution at 12% lignin and 4% PVA concentration.

[0295] The steady shear viscosity (as a function of shear rate) for dopes containing different lignins is shown in FIG. 7. The lignin type affects the dope viscosity. Dopes containing the two softwood kraft lignins display a lower viscosity than two hardwood ionoSolv lignins, which is likely a reflection of the different chemical and physical properties (for example average molecular weight, polydispersity and density) of lignins extracted from different feedstocks and with different processes. Shear thinning was observed for ionoSolv lignin-containing dope solutions. Especially for ionoSolv willow lignin, the shear thinning was apparent due to the high viscosity of the willow lignin dope across the tested shear rate range. For dopes containing Kraft lignin and PVA, significant shearing thinning behaviour was only observed when the shear rate increased above 400 s^{-1} and 100 s^{-1} for kraft lignin 1 and 2, respectively (FIG. 8).

[0296] The extrusion of dopes containing different types of technical lignins (75% lignin and 25% PVA) was performed to compare the spinnability and prove the robustness of the developed dope preparation and spinning method. The data show that homogenous dopes can be prepared from different lignins which can be spun continuously, then washed and dried. SEM images (FIG. 9) of the precursor fibres containing different lignins had smooth surfaces and circular cross-sections, indicating that a variety of lignins have good miscibility with PVA in the $[\text{DMBA}][\text{HSO}_4]_{60\%}/\text{water}_{40\%}$ dope solvent and the coagulation works well for all tested lignin types. The fibres containing willow lignin had the least smooth surface (small lumps) which is assigned to the undissolved lignin particles in the dope. However, this did not compromise the precursor fibres mechanical properties.

[0297] The mechanical properties of the lignin fibres were tested (Table 3). Fibres containing two types of Kraft lignin had smaller diameters which was due to the lower viscosity of their dopes. This resulted in a higher average tensile strength of the spun fibres. IonoSolv lignin-derived fibres had a higher stiffness and shorter elongation at failure. This

may be due to the better orientation of polymers as the shear rate during extrusion falls into the shearing thinning region.

TABLE 3

Diameters and mechanical properties of precursor fibres produced from various lignins				
Lignin source	Diameter (μm)	Tensile Strength (MPa)	Modulus (GPa)	Elongation at break %
Eucalyptus ionoSolv	120 ± 4	33 ± 1	4.5 ± 0.1	0.8 ± 0.1
Willow ionoSolv	140 ± 5	36 ± 2	4.3 ± 0.3	0.9 ± 0.1
Softwood Kraft 1	76 ± 1	48 ± 5	3.5 ± 0.5	1.7 ± 0.1
Softwood Kraft 2	80 ± 4	39 ± 4	3.4 ± 0.0	1.3 ± 0.2

Example 9. Optimisation of the Coagulant

[0298] The use of an aqueous 1M aqueous sodium sulphate (Na_2SO_4) solution and water for coagulating the lignin/PVA fibres was investigated. 1M sodium sulphate (Na_2SO_4) solution was initially used with the assumption that it would improve coagulation. Using water only as coagulant makes the fibre spinning process more sustainable, as the fibre washing step is eliminated and as it makes IL recycling simpler (no need to separate the sodium ions introduced by Na_2SO_4). When water was used as the coagulant, continuous and stable fibre spinning was observed. SEM imaging of a *Eucalyptus* lignin/PVA fibre with 75% lignin shows that the fibre has smooth surfaces and circular cross-section and there is no difference compared to the fibres spun in 1M Na_2SO_4 . Fibre coagulation in water was successful for the different types of lignin (ionoSolv and kraft) as well as different lignin content. The dimensions and mechanical properties of fibres coagulated in water and 1M Na_2SO_4 are shown in Table 4.

TABLE 4

Comparison of properties of lignin fibres coagulated in water and aqueous 1M sodium sulfate					
Lignin type (lignin content)	Coagulant	Diameter (μm)	Tensile Strength (MPa)	Modulus (GPa)	Elongation at break %
Eucalyptus ionoSolv lignin (75%)	1M Na_2SO_4	120 ± 4	33 ± 1	4.5 ± 0.1	0.8 ± 0.1
Eucalyptus ionoSolv lignin (75%)	H_2O	123 ± 12	30 ± 2	2.8 ± 0.2	1.2 ± 0.1
Eucalyptus ionoSolv lignin (82.5%)	1M Na_2SO_4	130 ± 3	22 ± 2	4.2 ± 0.1	0.5 ± 0.0
Eucalyptus ionoSolv lignin (82.5%)	H_2O	129 ± 3	22 ± 2	4.3 ± 0.1	0.5 ± 0.0
Softwood kraft lignin 1 (75%)	1M Na_2SO_4	76 ± 1	48 ± 5	3.6 ± 0.5	1.3 ± 0.2
Softwood kraft lignin 1 (75%)	H_2O	70 ± 2	37 ± 10	4.8 ± 0.3	1.2 ± 0.2

TABLE 4-continued

Comparison of properties of lignin fibres coagulated in water and aqueous 1M sodium sulfate					
Lignin type (lignin content)	Coagulant	Diameter (μm)	Tensile Strength (MPa)	Modulus (GPa)	Elongation at break %
Softwood kraft lignin 1 (75%) 24 h aged-dope	H ₂ O	60 $\pm$ 3	55 $\pm$ 4	5.1 $\pm$ 0.2	1.2 $\pm$ 0.1
Softwood kraft lignin 1 (75%) 24 h aged-dope + high extrusion rate	H ₂ O	94 $\pm$ 1	46 $\pm$ 6	4.7 $\pm$ 0.1	1.1 $\pm$ 0.1

*(higher extrusion rate = 1.5 mL/h)

[0299] The use of [DMBA][HSO₄]_{10%}/water_{90%} as a coagulant was also investigated with the goal of facilitating dope solvent recycling. The use of IL as part of the coagulant would reduce the energy required for solvent recycling, as this is achieved by evaporating H₂O and recovering the IL. Therefore, the presence of IL in the coagulant reduces the amount of H₂O that must be removed to reconstitute the dope solvent.

[0300] The spinning dope was prepared as described in Example 3 and consisted of lignin and PVA (weight ratio 3:1) dissolved in [DMBA][HSO₄]_{60%}/water_{40%}. When [DMBA][HSO₄]_{10%}/water_{90%} was used as a coagulant, long fibres formed which appeared to be uniform and flexible and could be picked up and hung for drying under tension. The fibres were not washed in water. The fibres did not shrink during the drying process but instead doubled in length under tension. The expansion was only stopped when the weight connected to the bottom end of the fibre reached the benchtop surface. After drying, the fibres were stable and did not expand further. A fibre that does not shrink during drying is desirable, as this will facilitate winding and hence continuous production of precursor fibres, which is key for industrial deployment.

[0301] The dimensions and mechanical properties of fibres coagulated in [DMBA][HSO₄]_{10%}/water_{90%} are:

Diameter (μm)	Tensile Strength [MPa]	Young's Modulus [GPa]	Elongation [%]
42.59 $\pm$ 6.72	13.62 $\pm$ 2.41	1.56 $\pm$ 0.32	0.66 $\pm$ 0.15

[0302] Optimisation of spinning into H₂O and 1M Na₂SO₄ coagulants is possible with regards to coagulation time and other variables, including a washing step. Hence, it might be possible to improve the fibre properties spun into [DMBA][HSO₄] and water mixtures by changing the parameters of the dope preparation, the fibre spinning and/or fibre washing steps.

Example 10. Carbon Yield of Lignin and Precursor Fibres

[0303] The key motivation of using lignin as the precursor for renewable carbon fibres is the high carbon content of lignin which results in high carbon yields. The carbon yield of different lignins and the respective precursor fibres with 75% lignin content and 25% PVA content (assumed to be equivalent to the lignin:PVA ratio in the spinning dope) was

estimated by employing thermogravimetric analysis (TGA). The predicted carbon yields are listed in Table 5. Interestingly, all precursor fibres resulted in carbon yields as high as the carbon yield of the lignins alone, although the lignin content was 75% in the precursor fibres and PVA on its own has a very low carbon yield. This is another indication of chemical reaction between lignin and PVA in acidic IL during the dope preparation. The measured carbon yields of precursor fibres (~40%) are higher than the carbon yield of previously reported fibres with the same lignin content obtaining by wet spinning from DMSO (Föllmer, M. et al. *Advanced Sustainable Systems* 2019). It should be noted that these DMSO based dopes were not homogenous.

TABLE 5

Carbon yield of the lignins used in this study and lignin PVA fibres (75%/25%).	
Carbon Yield (wt %)	
Eucalyptus ionoSolv lignin	43.2 $\pm$ 1.8
Eucalyptus ionoSolv lignin precursor fibres	40.7 $\pm$ 1.2
Willow ionoSolv lignin	40.2 $\pm$ 2.0
Willow ionoSolv lignin precursor fibres	40.4 $\pm$ 1.4
Softwood kraft lignin 1	39.1 $\pm$ 0.9
Softwood kraft lignin 1 precursor fibres	36.6 $\pm$ 2.8
Softwood kraft lignin 2	36.9 $\pm$ 2.6
Softwood kraft lignin 2 precursor fibres	35.7 $\pm$ 2.6
PVA	6.7 $\pm$ 0.2

Example 11. Carbonisation of Precursor Fibres

[0304] Carbonisation of lignin fibres was carried out using the conditions stated in example 5 (1000° C., heating rate 1° C./min, hold for 2 h) and after thermal stabilising fibres as stated in example 5 (in air at 250° C., heating rate 0.2° C./min from 100° C. to 250° C., hold for 1 h). SEM images of carbon fibres showed that the circular shape of the fibres was retained during carbonisation.

[0305] Characteristics of the carbon fibres are shown in Table 6.

TABLE 6

Characteristics of carbon fibres derived from precursor fibres (PF) produced using different lignins and lignin content and in with different coagulants.						
	Lignin content in PFs (%)	Coagulant	Diameter (μm)	Tensile Strength (MPa)	Modulus (GPa)	Elongation at break %
Eucalyptus ionoSolv lignin	82.5	1M Na ₂ SO ₄	91 $\pm$ 5	329 $\pm$ 63	36.3 $\pm$ 1.3	0.95 $\pm$ 0.19
Eucalyptus ionoSolv lignin	82.5	water	94 $\pm$ 8	282 $\pm$ 35	36.2 $\pm$ 2.2	0.85 $\pm$ 0.10
Eucalyptus ionoSolv lignin	75	1M Na ₂ SO ₄	80 $\pm$ 5	296 $\pm$ 70	39.9 $\pm$ 3.7	0.77 $\pm$ 0.21
Eucalyptus ionoSolv lignin	75	water	77 $\pm$ 13	291 $\pm$ 37	39.8 $\pm$ 3.1	0.57 $\pm$ 0.12
Willow ionoSolv lignin	75	1M Na ₂ SO ₄	100 $\pm$ 6	186 $\pm$ 35	33.5 $\pm$ 1.9	0.57 $\pm$ 0.01
Softwood Kraft lignin 1	75	1M Na ₂ SO ₄	54 $\pm$ 2	353 $\pm$ 19	39.6 $\pm$ 0.9	0.95 $\pm$ 0.09
Softwood Kraft lignin 2	75	1M Na ₂ SO ₄	57 $\pm$ 12	449 $\pm$ 16	40.1 $\pm$ 3.0	1.31 $\pm$ 0.15

Example 12. Properties of Thermally Stabilised and Carbonised Fibres

[0306] An aqueous poly(vinyl alcohol) solution was prepared and mixed with [DMBA][HSO₄]. Following the formation of a homogenous solution, LignoBoost lignin was added and the mixture left to stir until a homogenous solution was obtained. The composition of the dope is provided in Table 7.

TABLE 7

Composition of LignoBoost lignin dope for wet spinning of fibres.			
Solid loading/wt %		Solvent composition/%	
Lignin	Poly(vinyl alcohol)	[DMBA][HSO ₄]	DI H ₂ O
12	4	60	40
Actual % composition			
10	3	52	34

[0307] Extrusion of the lignin dope in a static deionised water coagulation bath at a rate of 2.5 mL hr⁻¹ using a 30-gauge needle produced continuous free flowing fibres. The extruded fibre was gently guided to a swift made from acetal (Delrin®) and left to air dry. The continuous fibre line collected approximately 6 m length of a single fibre which was limited by the length of the take-up swift.

[0308] The precursor fibre (unprocessed extruded fibre) underwent thermal stabilisation at 250° C. (thermostabilised fibre) and carbonisation between 900-2200° C. (carbonised fibre). The carbon yields of the precursor fibres and the thermostabilised fibres were estimated by TGA, presented in Table 8 alongside comparative values for precursor fibres made from polyacrylonitrile (PAN) and mesophase petroleum pitch. The lignin fibre mechanical properties differ between the precursor fibre, the thermostabilised fibre and the carbonised fibre, as illustrated in FIG. 10. For example, lignin-derived fibres show an increase in tensile strength after thermostabilisation. For carbon fibres it is desirable to have high tensile strength and high tensile modulus, whereas for textile fibres a lower modulus is desirable.

TABLE 8

Carbon yield for as-spun fibres and fibres thermally stabilised at 250° C.			
Fibre		Carbon yield %	Standard error $\pm$ %
Lignin-derived fibre	Precursor	30.25	0.25
	Thermostabilised	58.16	0.18
Precursor fibre	Poly(acrylonitrile)	50-60	—
	Mesophase pitch	80-85	—

Example 13. Fibre Spinning Using Higher MW PVA

[0309] A fibre spinning study was performed using a high molecular weight 88% hydrolysed PVA (Sigma Aldrich, Mw 146,000-186,000 g/mol) and softwood kraft lignin purchased from Sigma Aldrich.

[0310] The method described in Example 3 was used to prepare a spinning dope containing [DMBA][HSO₄]_{60%}/water_{40%} with a solid loading of 16% and a lignin/PVA ratio of 3:1.

[0311] The dope was cooled to room temperature before transferring the dope into a 1 mL plastic syringe (Injekt™-F Fine Dosage Syringe, B. Braun, Germany) and slowing pushing and pulling the plunger to expel any possible air bubbles in the dope. The dope was extruded using a syringe pump (LEGATOR 100, KD Scientific, US) at an extrusion rate between 0.1 and 0.4 mL/min into a coagulation bath containing 1M Na₂SO₄.

[0312] Fibres were drawn using a tweezer. Fibres successfully formed and were left to coagulate in the Na₂SO₄ coagulation bath for 60 s, then transferred for 30 s into a wash bath containing deionised water to wash the salt off the fibre surface.

[0313] The fibres were dried hanging overnight with a suitable weight loading of aluminium foil attached at the bottom end, which was used to stretch the fibres.

Example 14. Integrated Lignin Extraction and Lignin Fibre Spinning

Biomass Fractionation (Lignin Extraction into [DMBA][HSO₄])

[0314] The lignin extraction was carried out following the ionoSolv pretreatment procedure (Gschwend, F. J. V et al. J. Vis. Exp. 2016, 2016 (114), 4-9). 9 or 12 g (oven dry-basis, ODW) biomass was added into a 100 mL pressure tube (Ace

Glass, Vineland, NJ, USA, front sealing), followed by adding ca. 30 g [DMBA][HSO₄]_{83%}/water_{17%} to obtain a suspension of 30% or 40% biomass loading and 20% water content. The biomass and ionic liquid solution were mixed well with a vortex shaker (VWR) until all biomass particles were in contact with the IL. The pressure tubes were placed in a preheated oven for 1 h at 150° C. The mixture in the pressure tubes was cooled and transferred into a 500 mL glass bottle, followed by mixing with 180 g absolute ethanol (EtOH), shaken well and left to rest for 1 h at room temperature. After 1 h, the mixture was separated into a cellulose rich solid and liquid containing ionic liquid, ethanol and the dissolved lignin (liquor) using vacuum filtration. The cellulose was air-dried. The pulp was washed with EtOH three more times, followed by Soxhlet extraction in absolute ethanol for 24 h. The liquor was collected and the water and most of the EtOH was evaporated from the combined liquor fractions using a rotary evaporator. The lignin extraction was carried out in triplicate. Liquors obtained from 30% and 40% biomass loading lignin extraction are labelled as Liquor 30 and Liquor 40.

Compositional Analysis of Wood Biomass and Cellulose Pulp

[0315] Compositional analysis was carried out according to a published standard procedure by the National Renewable Energy Laboratory (NREL) (Sluiter, A. et al, *Natl. Renew. Energy Lab.* 2008, No. April 2008, 17). Extractives were removed from *Eucalyptus* biomass using EtOH with a Soxhlet extractor for 24 h and quantified by measuring the weight difference (oven-dried basis). Around 300 mg of air-dry extract-free biomass (oven-dried weight basis, sieved to 180-850 µm) or recovered pulp was weighed out into a 100 ml pressure tube (Ace Glass) and the exact weight recorded. 3 mL of 72% sulfuric acid (Fluka) were added, and the samples stirred with a Teflon stir rod and the pressure tubes placed into a preheated water bath at 30° C. The samples were stirred again every 10 min for one hour. They were then diluted with 84 mL distilled water and a lid added. The samples were autoclaved for 1 h at 121° C. (Sanyo Labo Autoclave ML5 3020 U) and left to cool to until the pressure tubes could be opened. The samples were filtered through filtering ceramic crucibles of a known weight. The filtrate was filled in two Falcon tubes (for acid soluble lignin content and sugar content determination) and the remaining black solid washed with distilled water. The crucibles were dried in a convection oven (VWR Venti-Line 115) at 105° C. for 24±2 h. They were placed in a desiccator for 15 min before their weight was recorded. The crucibles were placed into a muffle oven (Nabertherm+controller P 330) and ashed to constant weight at 575° C. They were again placed in a desiccator for 15 min before their weight was again recorded. The content of acid insoluble lignin (AIL) was determined according to Equation 1:

$$\% \text{ AIL} = \frac{W_{\text{crucible plus AIR}} - W_{\text{crucible plus ash}}}{\text{ODW}_{\text{sample}}} \cdot 100\% \quad (1)$$

[0316] where $W_{\text{crucible plus AIR}}$ is the weight of the oven-dried crucibles plus the acid insoluble residue, $W_{\text{crucible plus ash}}$ is the weight of the crucibles after ashing to constant temperature at 575° C.

[0317] The acid soluble lignin content (ASL) was determined by UV analysis of the autoclaving filtrate at 286 nm (Perkin Elmer Lambda 650 UV/Vis spectrometer). 200 µL sample and 800 µL D.I. water were added into a cuvette (dilution 1:4), mixed well and the absorption A recorded. The ASL was calculated according to Equation 2:

$$\% \text{ ASL} = \frac{A}{l \cdot \epsilon \cdot c} \cdot 100\% = \frac{A \cdot V_{\text{filtrate}}}{l \cdot \epsilon \cdot \text{ODW}_{\text{sample}}} \cdot 100\% \quad (2)$$

[0318] where A is the absorbance at 286 nm, l is the path length of the cuvette in cm (1 cm in this case), ϵ is the extinction coefficient (25 L/g cm), c is the concentration in mg/mL, ODW is the oven-dried weight of the sample in mg and V_{filtrate} is the volume of the filtrate in mL and equal to 86.73 mL.

[0319] Calcium carbonate was added to the remaining filtrate until the solution pH reached 5. The liquid was filtered through a 0.2 µm PTFE syringe filter and submitted to HPLC analysis for the determination of total sugar content (Shimadzu, Aminex HPX-97P from Bio rad, 300×7.8 mm, purified water as mobile phase at 0.6 mL/min, column temperature 85° C.). Calibration standards with concentrations of 0.1, 1, 2 and 4 mg/mL of glucose, xylose, mannose, arabinose and galactose were used. Sugar recovery standards were made as 10 mL aqueous solutions close to the expected sugar concentration of the samples and transferred to pressure tubes. 278 µL 72% sulfuric acid was added, the pressure tube closed and autoclaved and the sugar content determined as described above. Sugar recovery coefficient (SRC) and the sugar content of the analysed sample were determined according to Equation 3 and Equation 4, respectively:

$$\text{SRC} = \frac{c_{\text{HPLC}} \cdot V}{\text{initial weight}} \cdot 100 \quad (3)$$

$$\% \text{ Sugar} = \frac{c_{\text{HPLC}} \cdot V \cdot \text{corranhydro}}{\text{SRC} \cdot \text{ODW}_{\text{sample}}} \quad (4)$$

[0320] where c_{HPLC} is the sugar concentration detected by HPLC, V is the initial volume of the solution in mL (10.00 mL for the sugar recovery standards and 86.73 mL for the samples), initial weight is the mass of the sugars weighed in, corranhydro is the correction for the mass increase during hydrolysis of polymeric sugars obtained by dividing the molecular weight of one polymeric sugar by its monomeric weight (0.90 for C6 sugars glucose, galactose and mannose and 0.88 for C5 sugars xylose and arabinose) and ODW is the oven-dried weight of the sample in mg.

Determination of Lignin Loading in the Liquor

[0321] The lignin loading for the integrated spinning liquor, defined as lignin weight ratio percentage relative to the liquid fraction of liquor, including IL, water, ethanol and any additional unidentified solutes used for spinning was calculated based on the difference in the lignin content of the raw biomass and the lignin content in the ionosolv pulp (as determined by compositional analysis), the pulp yield (oven-dried weight basis) and the weight of the mass of liquor used to prepare the integrated spinning dope. Mass of the liquid

fraction is mass of liquor minus mass of lignin in that liquor). Equations used are shown below (equation 5, 6 and 7):

$$W_{\text{lignin(biomass)}} = (\% \text{ AIL} + \% \text{ ASL}) \cdot \text{ODWbiomass} \quad (5)$$

$$W_{\text{lignin(pulp)}} = (\% \text{ AIL} + \% \text{ ASL}) \cdot \text{ODWpulp} \quad (6)$$

$$\% \text{ Lignin (lq)} = \frac{W_{\text{lignin(biomass)}} - W_{\text{lignin(pulp)}}}{W_{\text{liquor}} - (W_{\text{lignin(biomass)}} - W_{\text{lignin(pulp)}})} \cdot 100\% \quad (7)$$

[0322] where $W_{\text{lignin(biomass)}}$ and $W_{\text{lignin(pulp)}}$ are the weight of lignin in the raw biomass and the *Eucalyptus* pulp, respectively; ODWbiomass and ODWpulp are the oven-dried weight of raw wood and pulp, respectively; % Lignin (lq) is the lignin concentration in weight percent in the liquor and W liquor is the weight of liquor.

[0323] The water content in the integrated spinning dope was determined using a coulometric Karl-Fischer titrator (Mettler Toledo). The ionic liquid and residual ethanol contents were determined using a ¹H-NMR spectrum of the liquor. The signals of the methyl group on ethanol (δH (400 MHz, DMSO-d₆)/ppm: 1.05, t) and signal of the methyl groups on the butyl chain of [DMBA][HSO₄] (δH (400 MHz, DMSO-d₆)/ppm: 0.90, t) were used for calculating of the molar ratio between IL and EtOH, which was converted to a weight ratio by multiplication with the molecular weight of each molecule.

[0324] where % IL (lq), % EtOH (lq) and % Water (lq) are IL, EtOH and water weight percentage in the liquor respectively. $W_{\% \text{ IL, EtOH}}$ is the weight ratio of IL and EtOH calculated from molar ratio obtained from the ¹H-NMR spectrum analysis.

Preparation of Integrated Spinning Dope Solution

[0325] A PVA aqueous solution of a certain concentration was prepared based on the desired lignin/PVA weight ratio. The investigated Lignin:PVA ratio in the fibres were 3:1 or 75:25 wt % and 4.71:1 or 82.5:17.5 wt %) and the final water content in the dope (20 wt %). Surprisingly, a lower water content was possible because of the presence of ethanol in the integrated spinning dope.

Example

[0326] After *Eucalyptus* wood fraction with a 40% biomass loading, a liquor with a 11% lignin loading and 0.7% water loading (numbers are the weight ratio percentage of lignin and water relative to the liquid fraction of liquor), which was used to prepare a dope with lignin:PVA=82.5:17.5 wt % and 20% water content by adding an 8.48% (w/w) aqueous solution of PVA. Using the same liquor to prepare a dope with lignin:PVA=75%: 25% and 20% water content, a PVA(aq) of 13.16% (w/w) was prepared.

[0327] To prepare the spinning dope, a PVA aqueous solution was mixed with the liquor at 60° C. while stirring at 700 rpm for 1 h inside a 10 mL round bottom flask with stir bar. The amount of PVA added to the liquor depends on the amount of liquor added in the round bottom flask. For example, to prepare a dope with polymer ratio of 3:1 (lignin:PVA=75%: 25%) and 20% water content, 1.00 g the liquor mentioned above was added, followed by adding 0.28 g of the 13.16% PVA(aq) solution.

Fibre Spinning

[0328] The dope solution was aspirated using a 1 mL plastic syringe (Normject™ Disposable) and air was carefully expelled. The dope solution (ca. 0.5 mL) was extruded into the coagulation bath (deionised water) which was placed on a rotational table at 3.5 RPM with a syringe pump (KDS LEGATO™ 100, KD Scientific, US) at an extrusion rate of 1.5 mL/h. For the spinning of fibres containing 75% lignin, a 27 g and 29 g needle (internal diameter 210 and 184 μm, respectively) was used. For the spinning of fibres containing 82.5% lignin, a 29 g needle (internal diameter 184 μm) was used. The bath was rotated at ca. 3 cm/s at the point of injection to provide extensional flow during fibre coagulation. The ratio of rotating speed at the point of injection to the injection speed was 2.5. The fibres were soaked for 2 minutes in the coagulation bath (water) and then picked up. Fibres with 75% lignin content were hung on a rod under tension from a known weight that was attached to the bottom end of the fibres (~0.5 mN) and fibres with 82.5% lignin content were hung under 0-0.1 mN tension. Fibres were dried overnight.

Results and Discussion

Lignin Extraction and Liquor Characterisation

[0329] Lignin was extracted into [DMBA][HSO₄] at 150° C. for 1 h, as described above. From the compositional analysis of raw biomass and recovered pulp, it was calculated that the delignification (describing lignin extraction efficiency) was 80.0% and 72.5% when the biomass loading was 30% and 40%, respectively (Table 9). Although more lignin could be extracted for 40% biomass loading, the delignification was reduced, which may be explained by reduced contact between biomass matrix and ionic liquid solution at higher biomass loading. The amount of lignin extracted in lignin/IL mixture (liquor) was calculated from the difference in lignin content on *Eucalyptus* raw biomass and recovered pulp. According to the calculation, lignin content was 8.8% and 9% in BL30 and BL40, respectively. Other compositions (IL, water and residual EtOH) are shown in Table 10. The liquors are homogenous according to the optical micrograph, with no undissolved particles observed, however, under the cross-polariser a few small crystalline birefringent particles were observed, which are likely to be cellulose crystals.

TABLE 9

Lignin content of eucalyptus raw biomass and the pulp, pulp yield after ionosolv fractionation and delignification			
	Lignin (%)	Pulp yield (%)	Delignification (%)
Eucalyptus	36.8 ± 0.1		
Eucalyptus pulp 30%	16.8 ± 0.9	45.6 ± 2.3	80.0 ± 2.1
Eucalyptus pulp 40%	19.9 ± 0.2	51.0 ± 0.5	72.5 ± 0.5

TABLE 10

Lignin and water weight ratio percentage in liquid fraction in liquor (including unknown solutes) and weight ratio of IL and EtOH in in liquid fraction in the liquor (including unknown solutes) of dried liquors obtained after fractionating Eucalyptus wood with [DMBA][HSO ₄] containing 20 wt % water				
Lignin extraction Biomass	Weight ratio percentage in the liquid fraction of liquor (including unknown solutes)		Weight ratio of IL and EtOH in the liquid fraction of liquor (including unknown solutes)	
	loading	lignin %	Water %	IL:EtOH (w:w)
BL30	30%	9.7 ± 0.4	0.9 ± 0.1	6.2 ± 0.3
BL40	40%	11.2 ± 0.3	0.7 ± 0.1	8.4 ± 2.3

[0330] The liquors without PVA demonstrated Newtonian flow behaviour (FIG. 11). At higher biomass loading (40%), the liquor had a slightly higher viscosity (ca. 0.6 Pa·s) than the liquor obtained with 30% biomass loading (0.4 Pa·s) in the shear rate region of 1-1000/s.

Spinning Dope Preparation

[0331] Lignin by itself cannot be wet-spun due to its low molecular weight and branched structure, therefore a strongly fibre-forming polymer (in this example, PVA) is added to the reconcentrated liquor in the same way as when preparing a dope from isolated lignin as in previous examples.

[0332] Ethanol was found to be present in the reconcentrated liquid, which had been added during the cellulose pulp washing step. 1H NMR spectra were used to quantify the ethanol content. A concentrated aqueous solution of PVA was prepared to achieve a final water content in the dope of ca. 20% upon addition (roughly 30% co-solvent content when including the ethanol in the liquor), which was different to the co-solvent content in the spinning method using isolated lignin (40 wt % water content). Subsequently, PVA aqueous solutions were prepared based on the known liquor composition and the desired lignin/PVA ratio in the fibres. When the lignin/PVA ratio was increased to 82.5/17.5 wt %, a more dilute PVA stock solution was prepared. The final dope compositions used for spinning are shown in Table 11.

TABLE 11

Weight ratio percentage of lignin and PVA in dopes relative to the dope solvent fraction (consisting IL, EtOH and unknown solutes), prepared from liquor BL30 and liquor BL40 (water content = ~22%, relative to the liquid fraction of dope).				
	Biomass	Desired lignin	Polymer loading	
	loading	content in fibre	lignin %	PVA %
Dope BL30_75%	30%	75%	8.1	2.7
Dope BL40_75%	40%	75%	8.6	2.9
Dope BL40_82.5%	40%	82.5%	8.6	1.8

Dope Rheology

[0333] The viscosity of integrated spinning dopes was measured and compared to the viscosity of liquor. Slightly higher viscosities in the shear rate range of 1-1000 s⁻¹ were observed for the integrated spinning dopes with a lignin/

PVA ratio of 75/25% compared to the reconcentrated liquor. Shear-thinning behaviour was observed for the dopes, whereas liquors without added PVA only exhibited Newtonian flow behaviour. Adding the PVA aqueous solution increases the water content, which is expected to reduce dope viscosity, while adding the co-polymer is expected to increase the viscosity. FIG. 11 shows that addition of the PVA aqueous solution overall increased the viscosity when the target polymer ratio was 75/25% (lignin/PVA), both for the 30% and the 40% biomass loading liquors, and shear thinning behaviour typical for spinning solutions was observed.

[0334] However, when the lignin/PVA ratio increased to 82.5/17.5%, the dope viscosity substantially reduced to 0.3 Pa·s, with the viscosity of dope being now lower than the viscosity of the reconcentrated liquor. The dope also appeared to behave like a Newtonian fluid at shear rates below 800 s⁻¹. This was due the reduced amount of PVA (1.8%) in the dope (2.7% in Dope BL30_75% and 2.9% Dope BL40_75%), as shown in Table 11.

Fibre Spinning and Characterisation

[0335] In general terms, extrusion pressure can influence coagulation speeds. Low pressure of extrusion, for example low viscosity of the dope, can result in faster coagulation. Extrusion parameters can be varied to optimise coagulation and fibre formation. For example, to increase extrusion pressure, one can use a needle with a thinner diameter or increase the dope viscosity.

[0336] In this example, for the dopes containing 75% lignin and 25% PVA, stable and continuous fibre spinning was achieved with an extrusion rate of 1.5 mL/h using a 27 G and 29 G needle (internal diameter 210 and 184 µm) at rotating speed of 3.5-9 RPM. Fibres could be picked up and dried under a small amount of weight (50 mg, ~0.5 mN). This was the case for the 30% or 40% biomass loading dopes.

[0337] Using a 29 G needle, 82.5% lignin fibres could be spun in a stable and continuous manner. The fibre spinning was repeated with dopes aged for 24 h and 48 h and with dopes heated and stirred for 6 h (instead of 1 h). Spinning of the aged dopes was also continuous and stable and an increased wet fibre strength was observed. These fibres could also hold a small weight (10 mg) during drying.

[0338] FIGS. 12a and 12b show that ageing and increased wet-fibre strength corresponded with an increase in integrated spinning dope viscosity, as expected. Mixing liquor with aqueous PVA for longer periods of time may further increase the dope viscosity and thus the wet-fibre strength.

Fibre Morphology

[0339] Fibres spun using the integrated process and containing 75% lignin had a uniform diameter and circular cross-section, which proves there was good coagulation during the spinning, as observed by SEM. The fibre surface was smooth, although small particles appeared on the surface which were not observed when spinning fibres with isolated *Eucalyptus* lignin.

Effect of Dope Preparation Time and Needle Size

[0340] As can be seen in Table 12, fibres spun from dopes prepared for 1 hour and 6 hours had a similar modulus and strain at break. Increasing the dope preparation time

appeared to increase the fibre diameter which was linked to the increase in dope viscosity. Interestingly, the average tensile strength of fibres increased, as well, when dope preparation time was increased. Without being bound by theory, this might be due to a higher shearing during extrusion of a higher viscosity dope.

TABLE 12

Tensile properties of precursor fibres (75% lignin) spun from the dopes mixed for different amounts of time. The dope was prepared from liquor BL30.					
Dope Prep. Time	Needle gauge	Diameter (μm)	Tensile Strength (MPa)	Young's Modulus (GPa)	Strain at break (%)
1 h	27	52.5 ± 3.9	41.4 ± 2.7	5.1 ± 0.1	0.9 ± 0.1
6 h	27	70.4 ± 2.7	45.5 ± 1.4	5.2 ± 0.0	1.0 ± 0.0
1 h	29	59.2 ± 3.6	43.6 ± 2.4	4.7 ± 0.1	1.0 ± 0.0
6 h	29	78.1 ± 3.4	47.5 ± 1.8	5.0 ± 0.3	1.0 ± 0.1

Effect of Biomass Loading

[0341] Lignin fibres prepared from liquors obtained from 30% and 40% biomass loading had similar fibre diameters and mechanical properties.

TABLE 13

Key mechanical properties of 75% lignin fibres spun from the liquors obtained with different biomass loading, dope was prepared for 1 h, spun from 29G needle				
Biomass Loading	Diameter (μm)	Tensile Strength (Mpa)	Young Modulus (Gpa)	Strain at break (%)
30%	59.2 ± 3.6	43.6 ± 2.4	4.7 ± 0.1	1.0 ± 0.0
40%	86.0 ± 1.5	43.4 ± 1.8	4.8 ± 0.1	1.0 ± 0.1

Carbon Yield

[0342] A high yield of carbon fibre is important for lowering the cost of carbon fibre production, a key requirement for increased adoption of sustainable carbon fibre composites.

[0343] This can be simulated on the small scale by thermogravimetric analysis. TGA showed that increasing the dope preparation time could increase the carbon yield. Fibres spun from the dope prepared from LQ30 heated for 1 h generated a ~30% carbon yield, while fibres spun from the dope heated for 6 h generated ~31% carbon yield. Fibres spun from the dope prepared from LQ40 has higher carbon yield, ~32% for fibres from 1 h-dope and 36% for fibres from 6 h-dope. For 82.5% lignin fibres, the carbon yield was around ~37%. During industrial carbon fibre production, precursor fibres are normally thermally stabilised in air prior to carbonisation to prevent fibres from melting or fusing during carbonisation. The carbon yield of thermal stabilised fibres was ~57% (Table 14).

TABLE 14

Carbon yield for precursor fibres and thermally stabilised fibres measured by TGA. Fibres were produced by integrated spinning of 30% and 40% biomass loading dopes.		
	Lignin content in fibre %	Carbon Yield (wt %)
BL-30 lignin fibre (1 h dope)	75	29.1 ± 2.6
BL-30 lignin fibre (6 h dope)	75	30.6 ± 0.6
BL-30 fibre (6 h dope) after thermal stabilisation	75	56.8
BL-40 lignin fibre (1 h dope)	75	32.2 ± 0.5
BL-40 lignin fibre (6 h dope)	75	35.7 ± 0.2
BL-40 lignin fibre (6 h dope)	82.5	36.6 ± 1.1

Example 15. Lignin Spinnability Study Using Different Ionic Liquids

[0344] The following ionic liquids were synthesised using methods known in the art:

Short name	Full chemical name
[HMim]Cl	1-methylimidazolium chloride
[DMBA]Cl	N,N-dimethylbutylammonium chloride
[TEA][HSO ₄]	Triethylammonium hydrogen sulphate
[DMBA][HSO ₄]	N,N-dimethylbutylammonium hydrogen sulfate
[BMim][HSO ₄]	1-Butyl-3-methylimidazolium hydrogen sulfate
[MBA][HSO ₄]	Methylbutylammonium hydrogen sulfate
[HBim][HSO ₄]	1-Butylimidazolium hydrogen sulfate
[DMBA][HCOO]	N,N-dimethylbutylammonium formate
[HMim][HCOO]	1-methylimidazolium formate
[DMBA][OAc]	N,N-dimethylbutylammonium acetate

[0345] Other chemicals used:

[0346] PVA, 88% hydrolysed (Kuraray Poval, No. 13-88, 100 kDa)

[0347] Kraft lignin: Softwood Kraft lignin (alkali), purchased from Sigma-Aldrich.

[0348] Ionosolv lignin: Extracted from *Eucalyptus* wood (hardwood) at 170° C., 1 h using [DMBA][HSO₄] with 20% water.

Solubility of Lignin in Ionic Liquid Water Mixtures

[0349] To study the spinnability of lignin-PVA dopes in the presence of different ionic liquids, the solubility of a hardwood (ionosolv lignin) and a softwood lignin (Kraft lignin) was initially examined in different ionic liquids water mixtures. Solubility of lignin in the dope solvent is an important pre-requisite for producing wet-spun lignin fibres. Preliminary studies found that PVA was soluble in a wide range of ionic liquid water mixtures (prepared by adding an aqueous stock solution of PVA), so a screening was not performed.

[0350] To determine lignin solubility, ionic liquid/H₂O mixtures with 20 wt % and 40 wt % H₂O were prepared, stirred and heated to 60° C. Lignin (12 wt %) was added and the mixture was stirred for 1 h, cooled to room temperature and the solubility of lignin was checked via polarizing optical microscopy. The solution was again stirred at 60° C.

for 1 h, and the solubility was checked a second time via polarizing optical microscopy.

Fibre Spinning

[0351] A basic spinnability test was performed with the ionic liquid water mixtures.

[0352] The dope preparation was performed as described in Example 3. Complete dissolution of the lignin within the dope was confirmed by polarizing optical microscopy. Fibre spinning was achieved by hand drawing with the use of a syringe pump. The dope was injected into a coagulation bath containing either aqueous 1M Na₂SO₄ or deionised (DI) water to form fibres. This was tested at extrusion rates between 0.01 mL/min and 0.1 mL/min.

[0353] To characterize the lignin fibres, they were dried overnight as described in Example 4. The use of a force to straighten the fibres during the drying process was applied for fibres strong enough to withstand this. Otherwise, they were dried without an additional force application.

Solubility of Lignin in Ionic Liquid Water Mixtures

[0354] The results of the solubility tests are provided in Table 15.

TABLE 15

Solubility of lignin in different ionic liquid water mixtures (✓ = soluble, x = not soluble).				
	Softwood Kraft lignin 20 wt % H ₂ O	Softwood Kraft lignin 40 wt % H ₂ O	Hardwood Ionosolv lignin 20 wt % H ₂ O	Hardwood Ionosolv lignin 40 wt % H ₂ O
[HMim]Cl	✓	x	✓	x
[DMBA]Cl	✓	✓	✓	✓
[TEA][HSO ₄]	✓	x	✓	x
[DMBA][HSO ₄]	✓	✓	✓	✓
[BMim][HSO ₄]	✓	✓	✓	✓
[MBA][HSO ₄]	✓	✓	✓	✓
[HBim][HSO ₄]	✓	✓	✓	✓
[DMBA][HCOO]	✓	x	✓	x
[HMim][HCOO]	✓	x	✓	x
[DMBA][OAc]	✓	✓	✓	✓

[0355] Some ionic liquids were only able to dissolve lignin at the lower H₂O content of 20 wt %. This can be explained by the anti-solvent character of water.

[0356] Comparing different ionic liquids, Lignin has high solubility in [DMBA]Cl, [DMBA][HSO₄], [BMim][HSO₄], [MBA][HSO₄] and [HBim][HSO₄], and dissolve the IL water mixture even in the presence of 40 wt % H₂O.

[0357] Previous reports have stated that the solubility of lignin is mostly determined by the anion of the ionic liquid, but the results show that the cation also has an influence.

[0358] The data collected suggest that the type of lignin has less influence on its solubility than ionic liquid composition and water content.

Fibre Spinning and Ionic Liquid Composition

[0359] Fibre spinning tests were carried out with a variety of IL/water dope solvents. The results are summarised in Table 16.

TABLE 16

Fibre spinning test results. Lignin and PVA were dissolved in ionic liquid water mixtures and extruded into deionised water or 1M aqueous sodium sulphate solution.				
	[MBA][HSO ₄] with 40 wt % H ₂ O	[BMim][HSO ₄] with 40 wt % H ₂ O	[DMBA][HSO ₄] with 40 wt % H ₂ O	[TEA][HSO ₄] with 30 wt % H ₂ O
SKL spun into aqueous Na ₂ SO ₄ :	✓ (fibres tear easily and are shorter)	x (agglomerates formed on the surface of the bath)	✓ (very good fibre formation)	✓ (thick and short fibres)
SKL spun into H ₂ O:	✓ (fibres tear easily and are shorter)	✓ (noncontinuous fibres)	✓ (very good fibre formation)	✓ (thick and short fibres)

TABLE 16-continued

Fibre spinning test results. Lignin and PVA were dissolved in ionic liquid water mixtures and extruded into deionised water or 1M aqueous sodium sulphate solution.					
HIL spun into aqueous Na ₂ SO ₄ :	✓ (fibres tear easily and are shorter)	x (agglomerates formed on the surface of the bath)	✓ (very good fibre formation)	x (dope too viscous)	
HIL spun into H ₂ O:	✓ (fibres tear easily and are shorter)	✓ (very thick and sticky fibres, not possible to pick them up)	✓ (very good fibre formation)	x (dope too viscous)	
	[HBim][HSO ₄] with 40 wt % H ₂ O	[HMim][HCOO] with 20 wt % H ₂ O	[DMBA][HCOO] with 20 wt % H ₂ O	[DMBA]Cl with 40 wt % H ₂ O	[DMBA][OAc] with 40 wt % H ₂ O
SKL spun into aqueous Na ₂ SO ₄ :	✓ (short, stiff non-continuous fibres formed)	✓ (continuous fibres formed)	✓ (continuous fibres formed)	x (agglomerates formed at the surface of the bath)	✓ (non-continuous fibres, partial formation of agglomerates near coagulant surface)
SKL spun into H ₂ O:	✓ (short, non-continuous fibres formed)	x (layer formed on bottom of the bath)	x (layer formed on bottom of the bath)	✓ (weak, continuous fibres formed)	✓ (non-continuous fibres, partial dissolution on coagulation bath floor)
HIL spun into aqueous Na ₂ SO ₄ :	✓ (continuous fibres formed with application of draw)	✓ (continuous fibres formed)	✓ (continuous fibres formed)	✓ (flattening of fibres upon touching coagulant surface)	✓ (more continuous fibres formed, some dispersion)
HIL spun into H ₂ O:	✓ (continuous fibres formed with application of draw)	x (layer formed at bottom of the bath)	x (layer formed at bottom of the bath)	✓ (flattening of fibres upon touching coagulant surface)	✓ (more continuous fibres formed, some dispersion)

(✓ = fibres formed, x = no fibres formed, SKL: softwood Kraft lignin, HIL: hardwood Ionosolv lignin).

[0360] A dope prepared using [DMBA][HSO₄] mixed with 40 wt % H₂O resulted in the formation of high-quality lignin PVA fibres, with Kraft lignin and IonoSolv lignin and using aqueous 1M Na₂SO₄ as well as deionised H₂O as coagulants. Whilst none of the other tested ionic liquids showed such good fibre formation under the conditions used for the screening, it should be noted that the protocol is optimised for [DMBA][HSO₄]. Improved fibre formation for the other ionic liquids may be obtained by optimisation of the spinning protocol, for example with a different pump system tolerating higher viscosities.

[0361] It will be appreciated that the above description is made by way of example and not a limitation of the scope of the appended claims, including any equivalents as included within the scope of the claims. Various modifications are possible and will be readily apparent to the skilled person in the art. Likewise, features of the described embodiments can be combined with any appropriate aspect described above and optional features of any one aspect can be combined with any other appropriate aspect.

1. A method for making fibres, the method comprising: providing a spinning dope comprising a dope solvent, lignin dissolved in the dope solvent and an additive polymer dissolved in the dope solvent, wherein the dope solvent comprises an ionic liquid and water; and

extruding the spinning dope into a coagulant to obtain one or more fibres.

2. The method of claim 1, wherein the water content in the dope solvent is at least 5 wt %, for example 5-40 wt %, preferably 10-40 wt %, more preferably 20-40 wt %.

3. The method of claim 1, wherein the spinning dope has a total loading of lignin and additive polymer of 6-60 wt %, 6-50 wt %, 6-40 wt %, 6-30 wt %, 8-23 wt %, 11-20 wt %, 10-20 wt %, 15-20 wt %, relative to the mass of mass of spinning dope, excluding the mass of lignin and additive polymer.

4. The method of claim 1, wherein the weight ratio of lignin to additive polymer in the spinning dope is at least 2:1 or at least 3:1, for example from 2:1 to 10:1, preferably from 3:1 to 9:1.

5. The method of claim 1, wherein lignin is present in the spinning dope at a loading of at least 5 wt %, relative to the total mass of dope solvent, preferably at a loading of 5-50 wt %, 5-40 wt %, 5-30 wt % or 5-15 wt %, relative to the mass of the total mass of dope solvent.

6. The method of claim 1, wherein the additive polymer loading is present in the spinning dope at a loading of 1-10 wt %, preferably 1-5 wt %, relative to the mass of spinning dope, excluding the mass of lignin and additive polymer.

7. The method of claim 1, wherein the additive polymer is selected from poly(vinyl alcohol) (PVA), polyvinyl

acetate, polyfurfuryl alcohol, polyacrylic acid, polyethylene oxide, polyethylene imine, poly(2-hydroxyethyl methacrylate), polyoxymethylene and mixtures thereof.

8. The method of claim 7, wherein the polymer is PVA.

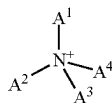
9. The method of claim 8, wherein the PVA is partially hydrolysed, having a degree of hydrolysis (DH) of 80-95%, preferably 85-90%, more preferably 86.7-88.7% hydrolysed.

10. The method of claim 1, wherein the weight average molecular weight (Mw) of the additive polymer is from 60 to 200 kDa, preferably 80 to 190 kDa.

11. The method of claim 1, wherein the ionic liquid comprises a cation and an anion selected from C₁₋₂₀ alkyl sulfate ([AlkylSO₄][−]), C₁₋₂₀ alkylsulfonate ([AlkylSO₃][−]), hydrogen sulfate ([HSO₄][−]), hydrogen sulfite ([HSO₃][−]), dihydrogen phosphate ([H₂PO₄][−]), hydrogen phosphate ([HPO₄]^{2−}), chloride (Cl[−]), bromide (Br[−]), trifluoromethanesulfonate ([OTf][−]), formate ([HCOO][−]) and acetate ([MeCO₂][−]); and/or

wherein the anion is selected from [HSO₄][−] and [HCOO][−]; and/or

wherein the ionic liquid comprises a cation and an anion, wherein the cation contains a nitrogen-containing heterocyclic moiety or wherein the cation is a cation of Formula I



wherein A¹ to A⁴ are each independently selected from H, an aliphatic, C₃₋₆ carbocycle, C₆₋₁₀ aryl, alkylaryl, and heteroaryl; and/or

wherein the ionic liquid is an [alkylammonium][HSO₄] or [alkylammonium][HCOO] ionic liquid.

12-14. (canceled)

15. The method of claim 1, wherein the ionic liquid is triethylammonium hydrogen sulfate [TEA][HSO₄], N,N-dimethylbutylammonium hydrogen sulfate [DMBA][HSO₄], diethylammonium hydrogen sulfate [DEA][HSO₄], N,N-dimethylethylammonium hydrogen sulfate ([DMEA][HSO₄]), diethanolammonium chloride [DEtOHA]Cl, 1-methylimidazolium hydrogen chloride [HMim]Cl, 1-ethyl-3-methylimidazolium chloride [EMim]Cl, and 1-ethyl-3-methylimidazolium trifluoromethanesulfonate [EMim][OTf], 1-butylimidazolium hydrogen sulfate ([HBim][HSO₄]), methylbutylammonium hydrogen sulfate ([MBA][HSO₄]), 1-methylimidazolium formate ([HMim][HCOO]), N,N-dimethylbutylammonium formate ([DMBA][HCOO]), N,N-dimethylethylammonium hydrogen sulfate

([DMEA][HSO₄]), 1-butyl-3-methylimidazolium hydrogen sulfate [BMim][HSO₄], or N-dimethylbutylammonium acetate ([DMBA][OAc]), or a mixture thereof, preferably wherein the ionic liquid is [DMBA][HSO₄].

16. The method of claim 1, wherein the dope solvent further comprises ethanol, optionally wherein the ethanol is present at 1-20 wt %, preferably 5-15 wt %, relative to the total weight of dope solvent.

17. The method of claim 1, wherein the coagulant comprises water; and/or

wherein the coagulant comprises water and ionic liquid, wherein the ionic liquid is present at no more than 60 wt %, no more than 30 wt %, or no more than 15 wt %, preferably 1-60 wt %, 1-30 wt % or 1-15 wt %, more preferably 5-10 wt %, relative to the total mass of coagulant; and/or

wherein the coagulant comprises water and sodium sulfate.

18-19. (canceled)

20. The method of claim 1, further comprising aging the spinning dope for at least 2 minutes, preferably at least 30 minutes prior to the extruding; and/or

further comprising drying the one or more fibres under mechanical tension; and/or

further comprising drying the one or more fibres by heating the one or more fibres in air, optionally at 150-300° C.; and/or

further comprising carbonising the one or more fibres to obtain carbon fibres, optionally wherein the carbonising comprises heating the one or more fibres to 800-3000° C., preferably 1200-1800° C., under an inert atmosphere.

21. The method of claim 1, wherein the providing the spinning dope comprises:

- contacting a lignocellulosic biomass comprising lignin and cellulose with a composition comprising the ionic liquid and water (preferably 5-40 wt % water), to dissolve the lignin and produce a cellulose pulp;
- separating the cellulose pulp to obtain a liquor comprising the ionic liquid, water and lignin; and
- combining the liquor with the additive polymer to obtain the spinning dope.

22. The method of claim 21, wherein the lignocellulosic biomass is contacted with the composition at 100-180° C.

23-24. (canceled)

25. The method of claim 24, further comprising weaving the fibres to form a fabric.

26. (canceled)

27. A fibre obtainable by the method of claim 1.

28. A fabric comprising one or more fibres according to claim 27.

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